

GEOTECHNICAL SOIL INVESTIGATION REPORT
FOR RESIDENTIAL BUILDING (4B + G + 10 + R)
FOR M/S. MDP DEVELOPMENT LIMITED
ON PLOT NO. JVC15AMRM004, AL BARSHA SOUTH FOURTH
DUBAI
UNITED ARAB EMIRATES

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FOR M/S. MDP DEVELOPMENT LIMITED.

ON PLOT NO. JVC15AMRM004, AL BARSHA SOUTH FOURTH

DUBAI, UNITED ARAB EMIRATES

Dear Sir,

It is our pleasure to submit the final geotechnical soil investigation report for the proposed Residential Building (4B + G + 10 + R), on Plot No. JVC15AMRM004, Al Barsha South Fourth, Dubai, United Arab Emirates.

Please note that the soil samples recovered from the site will remain in our storage facility for one calendar month, in case of any additional tests required. Our laboratory shall be informed if needed to keep the samples for extra time.

The recommendations given in this report are not in the scope of Emirates International Accreditation Centre (EIAC) accreditation and are based on the subsurface conditions encountered during the site investigation work and on the results of the field and laboratory testing on samples obtained from the Boreholes.

We hope that the information included cover all geotechnical issues related to the foundation of the project. For any further inquiry, please do not hesitate to contact us and we are looking forward to continued cooperation.

With Best Regards

For HASSAN AL AMIR SOIL ANALYSIS

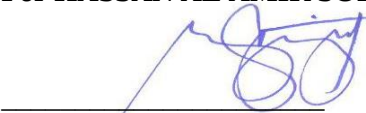

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1.0 INTRODUCTION

Hassan Al Amir Soil Analysis Laboratory was commissioned by M/s. MDP Development Limited to perform a geotechnical soil investigation works for the proposed construction of Residential Building (4B+G+10+R) for M/s. MDP Development Limited, on Plot No. JVC15AMRM004, at Al Barsha South Fourth, Dubai, United Arab Emirates.

The investigation has been based on a HASA proposal (QSI2022-0690, Rev02) dated 28th December 2022 and subsequent approval by M/s. MDP Development Limited as per Purchase Order No. SG2-23-0035 dated 04th January 2023. The work has been performed in general accordance with BS 5930: 2015, "Code of practice for site investigations".

2.0 AIMS AND SCOPE OF WORK

The aim of this study is to evaluate the subsurface conditions at the site and develop the essential criteria and parameters required to use in foundation design and construction for the proposed project. The investigation will provide the structure engineer with the main criteria for design of all structure members below the ground surface.

3.0 PROJECT AND SITE DESCRIPTION

The proposed project is the Residential Building (4B + G + 10 + R) and the structure is mainly made from reinforced concrete and different types of partitions. The site is located at Plot No. JVC15AMRM004, Al Barsha South Fourth, Dubai, United Arab Emirates. The plot was bounded by existing asphalt roads on Northwest and Southwest, building at the Northeast and another plot at the Southeast. The surface of the proposed plot was slightly undulating and is undeveloped. Plate 1 in appendix A of this report shows location plan and plate 2 shows a copy of the site plan as introduced from the client.

The topographic survey report which was provided by the client and reconnaissance of the area indicates that the plot is slightly undulating and its level ranges between +8.025m DMD & +8.433m DMD. [The average asphalt road edge level is +8.10m DMD for calculations.](#)

Table 1: Work Details of Boreholes

Borehole No.	Date Started	Date Finished	Depth, m	Elevation (m, DMD)
BH01	27/02/2023	28/02/2023	30.0	+8.35
BH02	22/02/2023	23/02/2023	40.0	+8.30
BH03	28/02/2023	01/03/2023	30.0	+8.15
BH04	27/02/2023	28/02/2023	30.0	+8.35
BH05	28/02/2023	01/03/2023	30.0	+8.25

4.0 GEOMORPHOLOGY, GEOLOGY AND WEATHER CONDITIONS

The typical Geomorphology of the site revealed an arid, low-lying setting and scarce vegetation. Geomorphological and palaeo-environmental investigations are integrated to provide a framework of environmental change for the Late Glacial and early mid Holocene periods against which the archaeology of the area can be set. The Late Glacial and earliest Holocene were characterized by intense aridity and accumulation of mega linear dunes driven by the Shamal winds.

The geology of the United Arab Emirates, and the Arabian Gulf area, has been substantially influenced by the deposition of marine sediments associated with numerous sea level changes during relatively recent geological time. With the exception of mountainous regions shared with Oman in the north- east, the country is relatively low-lying, with near surface geology dominated by Quaternary to late Pleistocene age, mobile aeolian dune sands, and sabkha/evaporite deposits.

The geologically stable Arabian Plate is separated from the unstable Iranian Fold Belt by the Arabian Gulf. It is believed that a tilting of the entire Arabian Plate occurred during the early Permian period, resulting in uplift in southern Yemen, and depression to the north- east. Crustal deformations and igneous intrusions occurred in the north- east as a result of this movement. Subsequent tectonic movements, peripheral to the folding of the Iranian Zagros Range, during the Plio-Pleistocene epoch, probably contributed to the formation of both the Arabian Gulf depression, and the mountainous regions shared by the United Arab Emirates and Oman in the north- east.

The offshore of the United Arab Emirates contains eight diapiric islands; Dalma, Zirkouh, Qarnain, Das, Sir Bani Yas, Arzana, Sir Abu Nu'air and Abu Musa. These islands owe their relief to the diapiric movement of salt which has pierced and deformed the overlying strata. The diapiric island has similar shapes, stratigraphic sequences, areal distribution of the identified stratigraphic units and general tectonic framework. With the exception of Das Island, the stratigraphic sequence on the surface of all islands consists, in ascending order, of:

- 1). Infracambrian to Cambrian (Hormouz Group) composed of igneous and metamorphic rocks, salt anhydrite, carbonate and clastic interbeds;
- 2). Miocene composed of sandstone, siltstone, shales, carbonate and evaporate interbeds; and,
- 3). Pliocene to recent sediments composed of mixed facies of clastic, carbonates and evaporates.

The structural configuration and the tectonic development of the Arabian Gulf Basin played an important role in the salt movement, which enhanced the formation and distribution of the islands, the timing of hydrocarbon generation, migration and entrapment in the surrounding fields. The United Arab Emirates, one of the world's

richest in oil reserves, has almost 200 billion barrels (Bbbl) of oil and 275 trillion cubic feet (TCF) of gas that is sourced mainly from the upper Jurassic and Lower to Middle Cretaceous formations and accumulated in carbonate reservoirs that range in age from upper Paleozoic to Oligo-Miocene. The geophysical and geological data revealed three trap geneses in the United Arab Emirates i.e. the Salt related, Basement related and Fold belt traps. Salt related oil fields of the United Arab Emirates offshore area are characterized by; (a) dome – shaped structures; (b) independent closures; (c) radial faults within the structures and (d) multi-step structural growth histories. Subtle turtle structures exist between the diapiric islands of the United Arab Emirates. These structures form fields at Hair Dalma and Dalma near Dalma Island, Mandous field near Sir Abu Nu'air Island, and Mubarek field near the Abu Musa Island. The quality of the carbonate reservoir in the salt related oil fields is attributed to the effects of the diapiric salt movement.

The tectonic development of these diapiric islands played a very important role in the mechanism of hydrocarbon entrapment. Most of the offshore oilfields in the United Arab Emirates are salt related traps and display a remarkable influence of the tectonic history of the area on the trapping mechanisms, including source rock deposition and maturation, timing of oil generation and migration, and trap formation. The mechanism of salt flow allowed the formation of high potential interdomal features known as turtle structures which are more abundant associated with salt diapirs that reach the surface. The Dalma and Hair Dalma oil fields and the Mandous oil field are turtle structures associated with Dalma and Sir Abu Nu'air salt diapirs, respectively.

Although surficial sabkha deposits are found throughout the coastal belt of the Arabian Gulf, and far inland in the western and southern parts of the United Arab Emirates, they are not particularly common in the DUBAI region. Sabkha is an Arabic name used to describe relatively fine grained, hypersaline, silt and sand deposits, which are commonly saturated with brine, and salt encrusted. These deposits form in natural evaporation pans where saline water, introduced by sea invasion or elevated saline groundwater, is able to accumulate at or near ground surface level, assisted by capillary action.

Conditions in DUBAI area essentially consist of a linear coastline dissected by channels or creeks. Superficial deposits consist of beach dune sands together with marine sands and silts. In addition, wind erosion, capillary action and evaporation has led to extensive Sabkha deposits in certain areas, notably around the creeks. These superficial deposits are underlain by alternating beds of calcarenite, carbonate sandstone, sands and cemented sands.

The site is located in DUBAI where a hot arid climate prevails. A hot arid climate is one where evaporation exceeds precipitation- such as rain, snow and dewfall. This climate regime produces characteristic hot desert terrains. Average annual rainfall may only be a few centimeters (even only a few millimeters in some parts)

which usually occurs seasonally and sometimes only from a single cloudburst. Summer shade temperatures are frequently in excess 40° C and humidity may be around 100% near the coast. The contrast between maximum night and day temperature and between night and day humidity is often great. Strong persistent winds are normally in many areas. This unfavorable climate imposes adverse conditions on the concrete structures, such as:

- High temperature and high seasonal changes
- High humidity and high change in relative humidity
- Strong drying winds
- Condensation at night
- Windborne salt – laden dust
- High solar radiation

5.0 FIELDWORK, EXPLORATION PROGRAM AND BORING TECHNIQUES

The geotechnical investigation works were performed in general accordance with the British Standard BS 5930: 2015 “Code of practice for site investigations” & BS1377:1990 Part 9. The borehole’s locations were set out in accordance with the instructions of the client’s representative and is shown in the boreholes location plan presented as Plate 3 in appendix A of this report.

A program of subsurface investigation, performed at the site from 08th August 2022 to 30th August 2022, include the following:

- Total of Five (05) Nos. Exploratory Boreholes with:
- Four (04) Boreholes to 30.0m Depths
- One (01) Borehole to 40.0m Depths
- Regular split-spoon sampling during Standard Penetration Tests (SPT) in soils
- Disturbed bulk soil sampling from cable percussion drill cuttings.
- Continues Coring in Rock and Core-able cemented soils.
- Occasional SPTs between core runs where recovery was low.
- Backfilling of boreholes with arising material
- Carrying out necessary laboratory tests such as particle size distribution, chemical analysis, unconfined compressive strength, etc.
- Performing engineering analysis of field and laboratory findings.
- Preparation of the final engineering report which includes the conclusions and recommendations for foundation design and construction.

The fieldwork was carried out under the close supervision of an experienced geotechnical engineer. The borehole logs are presented in appendix B of this report.

5.1 Standard Penetration Test (SPT)

A total of Five (05) boreholes were completed at the site to a proposed depth of between 30.0m and 40.0metres below existing ground level. Drilling was performed using a combination of rotary and cable percussion drilling rigs. The boreholes were advanced through soil deposits, including variably cemented sands, using cable percussive equipment and techniques. Rotary coring techniques were used in underlying rock, and core- able cemented sand formations.

Standard Penetration Tests (SPTs) were conducted regularly at 0.5m intervals up to 2.5m and thereafter 1.0m intervals up to the end of boring in accordance with the drilling procedure specified in BS 5930: 2015. SPTs were conducted in accordance with British Standard BS 1377 part 9: 1990 Cl. 3.3. See more details in appendix E of this report.

The test involves driving a 50mm external diameter thick-walled split spoon sampler into the bottom of the borehole with successive blows of a 63.5kg hammer falling freely through 760mm. The sampler is driven through 6 intervals of 75mm, and the number of blows required to penetrate each interval is recorded. The initial 150mm interval is intended to ensure “seating” of the sampler such that it penetrates beyond the zone of influence of any soil disturbance at the base of the borehole. The aggregate number of blows to drive the sampler over the final 300mm is termed the “N” value and is considered indicative of the in-situ relative soil density.

In very dense and/or cemented soil layers it is often not possible to ensure complete penetration of the SPT sampler, due to driving refusal, or the risk of damage to sampling equipment as a result of hard driving. The practical driving limit is of the order of 50 blows for the final 300mm penetration. Where driving resistance exceeds this limit, SPTs are usually terminated with incomplete penetration. In such cases the total 50 blows driven against the actual penetration depth is recorded on the borehole logs. A plot of SPT “N” values verses depths is presented in Appendix A, Plate 3.

Careful attention was paid to whether the groundwater level is present in the boreholes. Groundwater level was encountered in all boreholes. Water was added to assist cable percussive boring and to stabilize each hole during in-situ testing. Disturbed SPT and bulk soil samples collected during drilling were retained in sealed, labelled plastic jars.

5.2 Rock Sampling

Rotary coring in rock and core-able cemented soil was carried out using 1.5m, double tube, HWF rotary core barrel, producing a nominal core diameter of 76mm. Core runs were generally continued to the maximum 1.5m length although, runs were sometimes shortened to attempt to maximize core recovery where anticipated low.

All samples were transported to the HASA Lab storage facility for detailed logging and sample processing. Representative samples were selected for laboratory testing to assist with sample descriptions and determination of engineering properties. Individual borehole logs are presented in Appendix B.

5.3 Groundwater Table Measuring

The groundwater table was measured using Electric Contact Meter (Dip Meter), Type: KLL. Electric Contact Meters (Dip Meters) are portable, used for quick, easy, reliable instruments for measuring the water level and total depth in boreholes. See more details in appendix E of this report.

The groundwater table was encountered at a depth of approximately **+6.25m DMD ~ +6.15m DMD (i.e., 1.85m ~ 1.95m below the average adjacent asphalt road edge level)** at the time of finishing of boreholes. It has to be noticed that the groundwater table may be affected by seasonal and tidal changes as well as dewatering in the adjacent areas.

5.4 Backfilling of Boreholes

Each borehole was backfilled on completion with arising material 'sand' from bottom to top of the borehole, in order to avoid any future pollution of an aquifer. The site area was cleaned and cleared from any waste during investigation work.

6.0 SUBSURFACE CONDITIONS AND GROUNDWATER

The subsurface conditions as revealed from the boreholes indicated that:

- The site was underlain by a layer of medium dense to very dense, light brown to brown, silty, fine to medium SAND from existing ground level to 0.85m ~ 1.87m depths below the existing ground level, followed by extremely weak to very weak, light brown to brown, very thinly to medium bedded, fine to medium grained SANDSTONE to depths 12.5m ~ 13.75m.
- Below this stratum was extremely weak to very weak, light brown to light reddish brown, very thinly to medium bedded, fine to medium grained, slightly gypsiferous, SANDSTONE to depths 22.80m ~ 23.90m, followed by extremely weak to weak, light brown to brown, very thinly to medium bedded, fine to medium grained, slightly conglomeratic, slightly gypsiferous SANDSTONE to depths 27.90m ~ 33.00m.
- From depth 27.90m ~ 33.00m. very weak to weak, light brown to off-white, thinly to medium bedded, conglomeratic CALCISILTITE/ SILTSTONE was encountered to termination depths of 30.0m & 40.0m below the existing ground level.
- The groundwater table was encountered at a depth of approximately **+6.25m DMD ~ +6.15m DMD (i.e., 1.85m ~ 1.95m below the average adjacent asphalt road edge level)** at the time of finishing of boreholes.

The design subsurface profile summarized below in Table 2 is recommended for foundation design purposes. For more detailed, location specific descriptions, reference should be made to the borehole logs presented in Appendix B.

Table 2:- Recommended Design Subsurface Profile

Range (Below OGL)	Material Description
0.00m to 1.50m	Medium dense to very dense, light brown to brown, slightly silty, fine SAND.
1.50m to 13.00m	Extremely weak to very weak, light brown, fine to medium grained SANDSTONE.
13.00m to 23.0m	Extremely weak to very weak, reddish brown, fine to medium grained, slightly gypsiferous SANDSTONE.
23.00m to 28.0m	Very weak, reddish brown, slightly gypsiferous, conglomeratic SANDSTONE.
28.00m to 40.00m	Very weak to weak, light brown to off -white, fine grained CALCISILTITE/ SILTSTONE.

7.0 LABORATORY TESTING PROGRAM

A program of laboratory testing was performed on selected samples of soil, rock and groundwater obtained during the fieldwork. Test results have been used to assist with the classification and determination of relevant physical and mechanical properties of soil and rock samples. Chemical test results have been used to determine the corrosivity of soil and groundwater. In general, laboratory testing has been performed in accordance with British Standard procedures.

7.1 Particle Size Distribution

The particle size distribution was determined for a total of Four (04) soil samples in accordance with the wet sieving method described in BS 1377:Part2, 1990, Clause 9.2 "wet sieving method". Compliance with the standard, with respect to minimum sample quantity is dependent on the maximum significant grain size and the method of sampling.

In particular, for SPT samples, the quantity of soil available for testing is typically about 150g. This sample quantity is considered representative where grain sizes range up to 2mm (i.e. to coarse sand size). Where significant quantities of coarser particles are present, the particle size distribution obtained from SPT samples should be regarded as indicative only. It is particularly important to recognize the limitations of SPT samples where medium to coarse grained gravels and/or cobbles are present. Detailed presentations of results are preceded in Appendix C.

7.2 Unconfined Compressive Strength

A total of Forty-One (41) unconfined compressive strength (UCS) tests were conducted on selected rock core specimens in accordance with the methods outlined in ASTM D4543-19, ASTM D7012-14 & ASTM D2216-19.

Sample ends were carefully prepared by trimming with a circular saw, ensuring that each end was flat and plain. During testing, applied load and axial strain were monitored, enabling measurement of the UCS for each intact rock specimen. It is important to recognize that it is only possible to test intact samples, and that the test results do not therefore represent the degree of fracturing or weathering of the rock mass. These considerations are typically taken into account using empirical methods based on rock quality designation (RQD) measurements. Individual plots are presented in Appendix C.

7.3 Chemical Analysis

Chemical analyses were performed on One (01) soil sample and One (01) groundwater sample to determine sulphate content, chloride content, and pH levels. These analyses were performed in accordance with the BS 1377: Part 3:1990: Clause 5.3: Amd. 9028:96 methods for testing water soluble sulphate (expressed as 2:1 water: soil extract) and chloride contents in accordance with the BS 1377: Part 3:1990: Clause 7.2: Amd. 9028: 96. The pH levels were determined in accordance with the BS 1377: Part 3: Clause9: 1990, Amd. 9028: 96. The chemical test results are summarized in Appendix C.

8.0 DISCUSSION AND RECOMMENDATIONS

The purpose of the investigation was to evaluate the subsurface conditions with respect to the design of foundations for the proposed development. It is understood from the information provided by the client that it is proposed to construct Residential Building (4B + G + 10 + R) on this site, to be set on raft foundation. **The average adjacent asphalt road edge level is +8.10m DMD for calculations.**

The observed sub-surface conditions are such that relatively high load bearing soils are present below the influence zone of the formation level. Accordingly, Basement Raft type of foundation with corresponding estimated orders of settlement for a rigid **raft foundation to be set at approximately -4.05m DMD ~ -7.75m DMD (i.e., 12.15m ~ 15.85m depth below the average asphalt road edge level) for the building with four (04) levels of basement** is considered appropriate for the proposed main building, underground water tank and lift pit structures. It is further recommended that below the foundation the site should be properly leveled and then thoroughly compacted.

8.1 Allowable Bearing Pressure for Raft Foundation

Allowable bearing pressures are dependent on the shear strength of the soil and the tolerance of the proposed structure to settlement. In granular soils, the latter factor is normally more critical. Settlements in such deposits normally take place during construction and initial loading but, where more silty or cohesive materials exist; there could be a degree of time dependent consolidation.

Based on the Standard Penetration Tests (SPTs), Unconfined Compressive Strength (UCS) tests and careful assessment of soil and rock cores samples, it is recommended that following allowable bearing pressure, with corresponding estimated orders of settlement, are given below in Table 1 for a rigid raft foundation. These recommendations are based on the raft, being set at depths ranging from **-4.05m DMD ~ -7.75m DMD (i.e. 12.15m ~ 15.85m depth below the average asphalt road edge level) for the building with four (04) basement** on properly level and compacted surface.

Foundation Level, m, DMD/BRL	Net Allowable Bearing Capacity, kN/m ²	Allowable Settlement, mm	Modulus of Sub-grade Reaction, kN/m ³	Type of Foundation And Footing width
-4.05 ~ -4.75/ -12.15 ~ -12.85	600	50	36,000	Raft Foundation for Main Building Up to width 38.0m
-6.25 ~ -7.75/ 14.35 ~ 15.85	625	50	37,500	Raft Foundation for Main Building Up to width 38.0m

With loads of the magnitudes quoted above, total settlement is calculated as being less than the generally accepted tolerance of 50mm for raft, with differential movements being minimal. The bearing pressures for raft foundations were calculated by Meyerhof equations 4-13 and 10.1 given below;

$$q_a = N / F2(\Delta H_a/25)*K_d$$

Where $K_d = 1+0.33D/B$

ΔH_a = allowable settlement such as 25, 30,35,40,45 and 50mm etc.

Note: For Mat/ Raft foundations the ratio $((B+F3)/B)^2 \approx 1.0$ and is neglected.

The following shall be considered:

- Bottom of excavation must be compacted to at least 95% of the modified proctor maximum dry density of soil using a heavy compactor roller (20 tons or more). The compacted surface should then be protected by immediately covering it with a layer of concrete blinding. In case of excavation through rock strata, the excavated surface should be clean and clear from any loose material or debris.
- When excavating to foundation level, care should be taken not to disturb strata underlying formation level otherwise settlements in excess of those quoted above could occur. If any disturbance does occur, or **if any soft spots or gypsiferous materials are encountered, the material should be removed and replaced by selected approved granular fill**, which should be properly compacted before the foundations are constructed.

8.2 Groundwater Control/ Dewatering and Excavation for Foundations

Groundwater was encountered at depths of **+6.25m DMD ~ +6.15m DMD (i.e. 1.85m ~ 1.95m below the average adjacent asphalt road edge level)**. The building with four (04) levels of basement is considered to be set at around **-4.05m DMD ~ -7.75m DMD (i.e., 12.15m ~ 15.85m depth below the average asphalt road edge level)**, therefore **DEWATERING WILL BE REQUIRED**. Specialist dewatering contractors should be sought with regard to the suitability and economy of their proprietary systems.

The dewatering should remain in operation until the weight of the structure is greater than the uplift pressure exerted by the groundwater returning to its natural level. The waterproofing system adopted should be capable of withstanding the hydrostatic head that will be effective once the water table has returned to its natural condition. To ensure a dry, firm base to the excavation, it is therefore considered that the draw down level should be maintained at around 1.0m below formation level.

It should be noted, however, that groundwater levels are subject to variation caused by tidal and weather seasonal variations and by changes of local drainage and or pumping conditions. The levels may at time be significantly different to those measured during the investigation.

The following will be considered:

- The excavation works should be carried out in accordance with good construction practice, such as BS 6031: 1981 “Code of Practice for Earthworks” or a similar recognized standard.
- Temporary or permanent retaining structures may be constructed around the building line.

During the excavation of the plot, vertical sides will be required to the excavation. Sheet piles could be used for this purpose, or king piles with suitable infill panels. Considering the size of the excavation, it is considered that tie back anchors may be required in order to limit the deflection of the walls. If anchors are required, they should be designed to a standard, such as British Standard, BS 8081, “Code of practice for ground anchorages”, taking into account the aggressive nature of the soils and groundwater into which they will be placed. Alternatively, a permanent support system could be considered, such as secant piles or diaphragm walling, which could then be incorporated into the walls of the basement.

8.3 Backfill Materials & Compaction

The following notes must be considered as appropriate when applicable:

- The bottom of excavation/excavation surface must be compacted to at least 95% of the modified proctor maximum dry density of soil using a heavy compactor roller.
- Backfill material should not contain any roots, construction debris, or deleterious material.

Recommended engineering fill material shall comply with the following specifications:

Type of soil	:	Inorganic and free draining with no more than trace amounts of organic matter, gypsum or other salts.
Particle size	:	50mm max.
Fines content	:	15% max.
Gradation	:	Well graded.

- Structural backfill should be performed in horizontal layers of thickness 150mm to 250mm. Compaction should be performed in each layer to reach the required degree mentioned in this report. This is applicable for slabs on grade and asphalt pavements as well.
- The excavated materials cannot be used for structural fill without testing but can be used as common fill provided it is not supporting any structure, i.e. around the building or somewhere else.

8.4 Excavation Side Slopes and / or Lateral Support System

Unsupported temporary excavation walls and slopes should be conducted at gradients that secure their stability. Unsupported temporary excavations in the different strata present at the site should consider the following gradients:

- In the very loose soils, slopes should have maximum gradients of 1V: 2H;
- In the medium dense to dense soils, slopes should have maximum gradients of 1V:1.5H;
- In weak bed rock, slopes should have maximum gradients of 1V:1H.
- Slopes for more than 3.0m deep excavations shall be designed and supervised by an engineer on a case-by-case basis.

In case the above-mentioned gradients are not feasible due to the site-specific restrictions steeper slopes can be adopted considering appropriate slope stabilization measures.

Vertical unsupported excavation walls in the bedrock can be achieved; however, the maximum stable height should be determined for each particular case, taking into consideration the thickness of the topping soil, expected surcharge loads, geo-mechanical properties of the rock mass among other design parameters.

8.5 Earth Pressure, Soil Parameters & Retaining Structures

As discussed above, development at the site will involve excavation up to approximately **-4.05m DMD ~ -7.75m DMD (i.e., 12.15m ~ 15.85m depth below the average asphalt road edge level)** for setting up the formation level is required. Retaining walls will therefore be required as part of the structure and possibly also as temporary work. The design parameters for retaining structures are given below in Table 3.

Table 3: Recommended Design Parameters for Retaining Structures / Shoring Design

Depth (m OGL)	Stratum	Bulk Density (Mg/m ³)	Submerged Density (Mg/m ³)	Internal Friction Phi Φ (degree)	Unit Weight γ KN/m ³	Cohesion Intercept (C') (kN/m ²)	Coefficient of Lateral Earth Pressure		
							Ko	Ka	Kp
0.0 to 1.50	Medium dense to very dense SAND.	1.75	0.75	33	17.50	0	0.46	0.29	3.39
1.50 to 13.00	Extremely weak to very weak SANDSTONE.	1.85	0.85	34	18.50	30	0.44	0.28	3.54
13.00 to 23.00	Extremely weak to very weak, gypsiferous SANDSTONE	1.90	0.90	32	19.00	30	0.47	0.31	3.25
23.00 to 28.00	Extremely weak to very weak, gypsiferous, conglomeratic SANDSTONE	2.00	1.00	33	20.00	30	0.46	0.29	3.39
28.00 to 40.0	Very weak to weak CALCISILTITE.	2.00	1.00	32	20.00	30	0.47	0.31	3.25

In the above table Ko has been determined using Bishop's procedure, with Rankine's method being used to calculate both Ka and Kp.

8.6 Liquefaction Analysis

Soil liquefaction is a phenomenon in which saturated loose cohesion less soil are subjected to a temporary, but essentially total loss of strength induced by severe earthquakes. Significant factors known to affect the liquefaction potential of these soil are the characteristic of the material, such as the grain size distribution and relative: the initial stresses or monotonic shear stresses that could even trigger flow liquefaction by the generation of excess pore water pressure and strain softening; and the characteristics of the earthquake in, such as the intensity and duration the of ground shaking.

In this site, the subsurface materials are medium dense to dense, grading to very dense, light brown to brown, silty, fine to medium, SAND from existing ground level to top of the rock encountered at 0.85m ~ 1.87m to termination depths 30.0m & 40.0m depths below existing round level.

Based on the subsurface materials encountered and the results of the Standard Penetration Tests (SPT) and Unconfined Compressive Strength (UCS) performed in Five (05) boreholes drilled at the site, **The SOIL WILL NOT LIQUEFY** due to earthquake considered.

8.7 Concrete Design for Foundation Purposes

Based on the chemical tests performed in our laboratory shown in appendix C of this report and BRE Special Digest 1, 2005 "Concrete in Aggressive Ground" the following is noted:

Maximum Sulphate Content		Maximum Chloride Content		Lowest pH	Design Sulphate Class	Aggressive Chemical Environment for Concrete (ACEC) Class
2:1 water/soil extract, SO ₄ , g/l	Groundwater, g/l	Acid soluble extract, Cl, %	Groundwater, g/l			
0.44	2.07	0.04	0.03	7.3	DS-4	AC-4

The above classification does not reflect the significant of chloride content in concrete surrounding, so the exposure conditions shall be studied based on CIRIA Special Publication 31 (The CIRIA Guide to Concrete Construction in the Gulf Region published by the Construction Industry Research and Information Association, London 1984) shown in appendix E of this report which considers both sulphate and chloride contamination of soils and groundwater in various exposure conditions and is not restricted just to sub-surface concrete.

Based on the chemical tests results and CIRIA Special Publication 31, it is considered that this site would be classed as having significant contamination of sulphate salts and moderate contamination of chloride salts. On this basis it is considered that Exposure Condition d(iv) would apply, for which a cement containing at least 3.5% C₃A but not more than 9% C₃A is preferred, however, an ASTM type II cement (MSRPC) is recommended to be used in a mix meeting the following criteria:

Minimum cement content for 20mm aggregates	320 to 400 kg/m ³
Maximum water: cement ratio	0.50 to 0.42
Minimum cover for reinforcement	40mm to 50mm

In addition, it is recommended that the surface of all concrete in contact with the soil should be covered with a suitable waterproof membrane or coating. However, prior to finalizing and adopting a mix design, appropriate consultation with the structural engineer is advised.

8.8 Seismic Zone & Soil Profile

As no geophysical testing (i.e., cross-hole or down-hole seismic testing) was performed at the site, no analysis was made in the report. It should be noted that the below reported values are considered to be tentative and based on the results of the drilled boreholes. Only geophysical logging of boreholes can give exact value of the shear wave velocity.

Based on the results of the investigation and with reference to the **American Society of Civil Engineers (ASCE)-7-16, Uniform Building Code (UBC) 1997 & Dubai Building Code 2021 edition**, the seismic site class and other parameters for the proposed project site is given as follows:

Description	Parameter	Recommended Value	Reference
Mapped Spectral Acceleration for short period	S _s , g	0.33	Dubai Building Code Table F.20
Mapped Spectral Acceleration for 1-sec period	S ₁ , g	0.11	
Long-Period Transition Period	TL, s	24	
Site Class/ Soil Profile Type	Extremely weak to very weak SANDSTONE	C/ S _C	ASCE7-16, UBC 1997 Table 20.3-1
Short-Period Site Coefficient	F _a	1.2	ASCE7-16 Table 11.4-1 & Table 11.4-2
Long-Period Site Coefficient	F _v	1.5	
Peak Ground Acceleration	PGA	0.13	Dubai Building Code Table F.20
Seismic Coefficient	C _a	0.28	UBC 1997
Seismic Coefficient	C _v	0.40	UBC 1997
Seismic Zone Factor	Z	0.20	DM Circular No. 191

9.0 GENERAL COMMENTS AND LIMITATIONS

The following points must be considered in conjunction with this report:

- As mentioned in the report the boreholes made were 100/150mm in diameter. The information received from the limited number and the diameter of boreholes may not represent the entire site and may not reveal all the weak layers or conditions especially when they are localized. Hence no responsibility can be borne for conditions not revealed by boreholes made on the site. In case of any such findings, we shall be contacted immediately to arrange a site visit by a geotechnical engineer to make an onsite study of such conditions after which recommendations if it is deemed necessary will be revised.
- The borehole logs and related information represent subsurface conditions only at the specific locations and times where sampling was conducted. Any lines designating the changes between soil and/ or rock layers represent approximate boundaries. The transition between deposits / strata may be gradual or may occur between recovered samples.
- There may be conditions pertaining to the site which have not been in to account due to the limited number of boreholes.
- Precautions should be taken to prevent any ingress of water into more silty horizons both during and after construction, otherwise settlement in excess of those quoted in this report could occur.
- At the time of construction, it shall be made sure that the excavation levels for foundations are at least 0.3m above the groundwater table so that compaction process at the excavated level can be properly carried out to the required degree mentioned in our report. In case of any difficulty in achieving proper compaction, a 250mm layer thick granular fill material shall be placed and compacted to the required degree of compaction.
- While carrying out dewatering, it shall be ensured that no excessive settlement is caused to the nearby structures and the process shall be carried out in stages. Moreover, suitable precautions have to be taken, especially in areas with weak strata.
- Where the construction area levels are less than the proposed foundation level, filling below the foundation level shall be carried out in layers not exceeding 250mm and each layer shall be compacted to at least 95% of the modified proctor maximum dry density of the soil in case of fine material or to 98% in case of granular fill (road base). The compaction percentages attained shall be confirmed by carrying out in situ density tests for each compacted layer. For backfills up to 1.0m, plate load test shall be carried out to confirm the efficiency of the backfill. In case of the backfill exceeds 1.0m, cone penetration test (CPT) shall be carried out to confirm the same.
- Before pouring the concrete, the ground surface at foundation level should be free of any loose stone fragments and dirt.

- In case of any hard strata like the sandstone, siltstone or other rock types, use of a special excavators for excavations of such strata shall be considered.
- For excavations deeper than 2.0m, suitable side protections have to be ensured so that the excavation shall not pose a threat to the personnel working on the site or cause any damage to nearby existing buildings or roads.
- In case there is an existing structure, which is waiting for demolition, and the soil investigation has to be carried out outside the existing structure it is advisable to sink one or two more boreholes after demolition to confirm the bearing capacity.
- Suitable slopes shall be used to drain storm and wastewater away from the proposed building.
- The recommendations given in this report need not be considered as a final and binding. These recommendations may be altered (where the soil parameters permit) depending on the design requirements and by agreement with the geotechnical engineer.
- All earth works including bottom of excavation inspection, site filling and the specified tests prior to and during construction shall be carried out under the supervision of a qualified geotechnical engineer.
- The tests and experiments for the quality control on soil, construction materials, and concrete must be considered, followed up and approved by HASA laboratory.
- HASA laboratory shall be contacted to carry out compaction or other tests if recommended in our report at random locations selected by us in which case only a letter of confirmation can be issued for the safe bearing capacity and foundation levels.

Prepared By

Approved By

Engr. Mary Genifer C. Sambade
Sr. Geotechnical Engineer

Engr. M. Khursheed Aziz
Laboratory Manager



APPENDIX A

LOCATION PLAN

SITE PLAN

BOREHOLES LOCATION PLAN

SPT "N" VALUES VS DEPTH

GEOTECHNICAL CROSS SECTION PROFILE

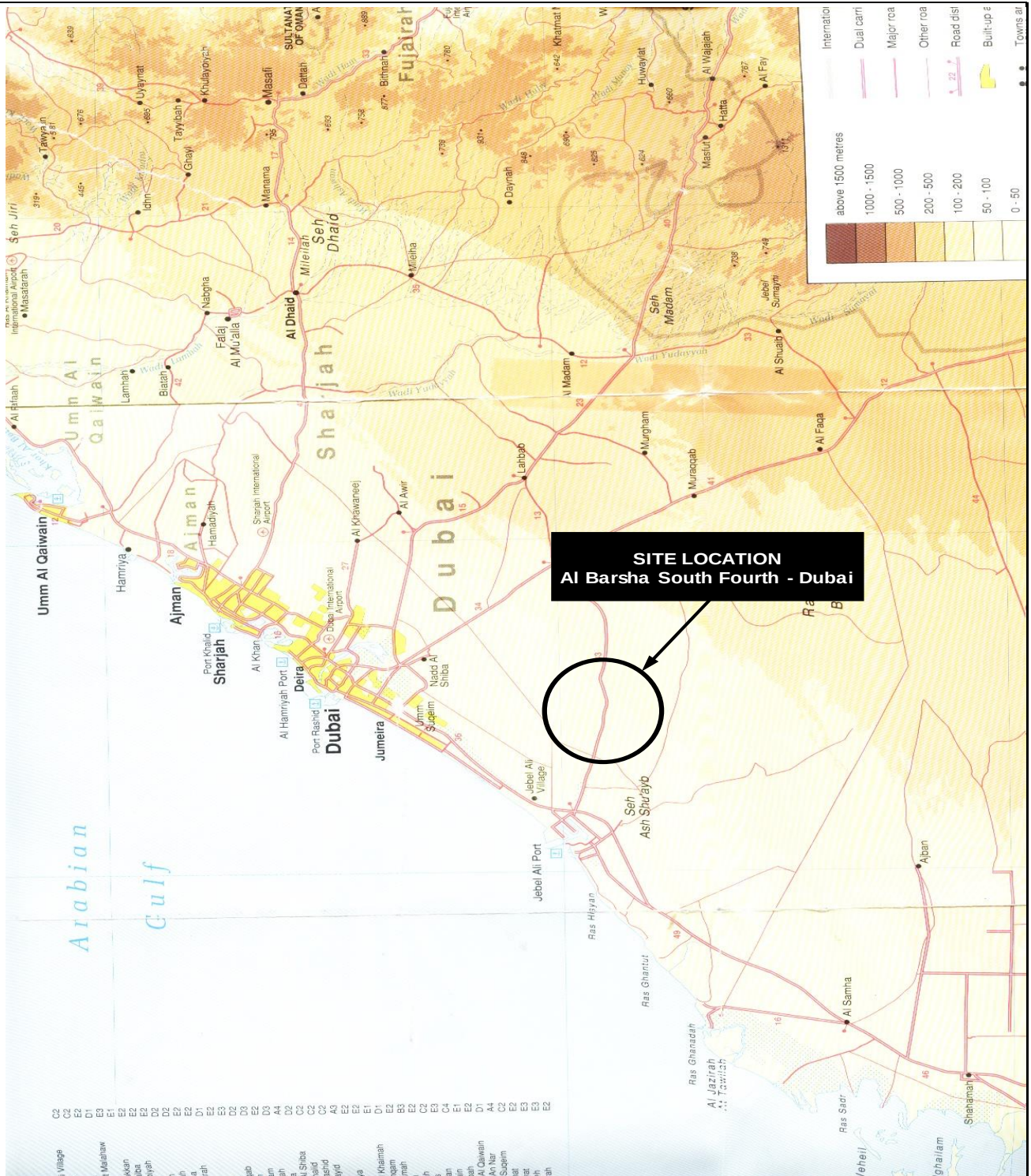
PLATE 1

PLATE 2

PLATE 3

PLATE 4

PLATE 5



JOB REF : SIR2023-0018

DATE: 14-Mar-23

SITE: Building (4B + G + 10 + R)
JVC15AMRM004, Al Barsha South Fourth
Dubai
United Arab Emirates

CLIENT: M/s. MDP Development Limited

TITLE: Location Plan

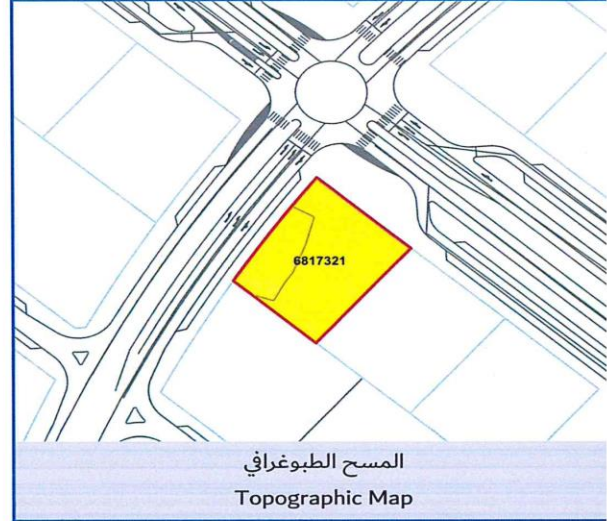
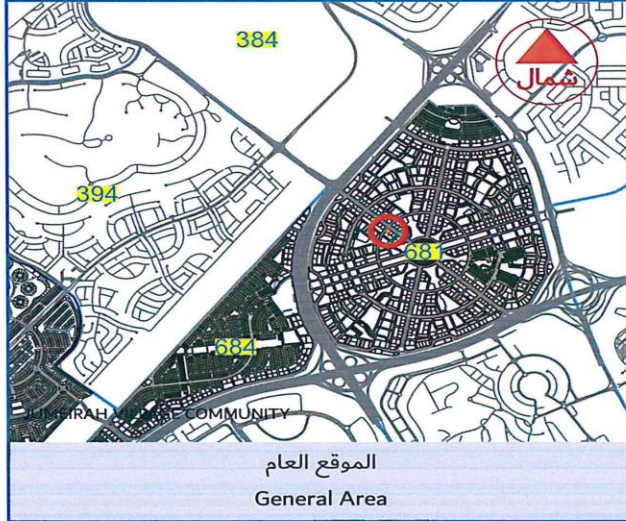
PLATE No: A1.1

Plate 1.1 General Location Plan

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Tel 04 – 341 6086 Fax 04 – 341 6096 - P.O. Box 510.

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تليفون 04 – 341 6086 فاكس 04 – 341 6096 – ص.ب. 510

Website: www.soilanalysis.hagalamir.com / Email: hasalab@hagalamir.com, hasa.lab1@gmail.com



Mixed use (Residential + Retail + Commercial)

الاستعمال
Land Use

(Refer to Trakhees Regulations & Dubai Building Code in parallel with the master developer's DCR)

المواقف
Parking

(Refer to Trakhees Regulations & Dubai Building Code in parallel with the master developer's DCR)

الارتداد
Setback

(Refer to Trakhees Regulations & Dubai Building Code in parallel with the master developer's DCR)

ارتفاع المبنى
Building Height

(Refer to Trakhees Regulations & Dubai Building Code in parallel with the master developer's DCR)

غطية قطعة الأرض
Plot Coverage

- The Plot is subject to the Master Community Declaration and Development Control Regulations of Jumeirah Village
- Infrastructure in Jumeirah Village is not jointly owned property and is the sole property of Jumeirah Village LLC. Infrastructure encompasses the following:
 - The roads, turns, crossroad, corridors, pavement edges, drainage sewers, island separating the road, arch bridges and drainage systems and their relevant parts.
 - Lakes, pools, canals, parks, fountains, water features and other watercourse including and all relevant equipment;
 - Green bodies, public areas and playgrounds;
 - Wires, cables, pipelines, drainage sewers, canals, machinery and equipment utilized for providing units or common parts with utilities; and
 - Instrumentations of extending or providing services for the utilities considered as allocated for common usage by the unit owners and occupants

ملاحظات
Remarks

معلومات التخطيطية لقطعة الأرض صالحة طالما أنها لا تتعارض مع اللوائح التخطيطية لتراخيص / المطور، يجب على المالك من خلال المطور التحقق من هذه المعلومات والمنطقة المعنية قبل الشراء أو البيع أو التأجير
The plot planning information is valid as long as it does not contradict with Trakhees/developer planning regulations. The client through developer should verify this information and the concerned area prior to purchasing, selling or leasing



SITE: Building (4B + G + 10 + R)
JVC15AMRM004, Al Barsha South Fourth
Dubai
United Arab Emirates

TITLE: Location Plan

JOB REF : SIR2023-0018

DATE: 14-Mar-23

CLIENT: M/s. MDP Development Limited

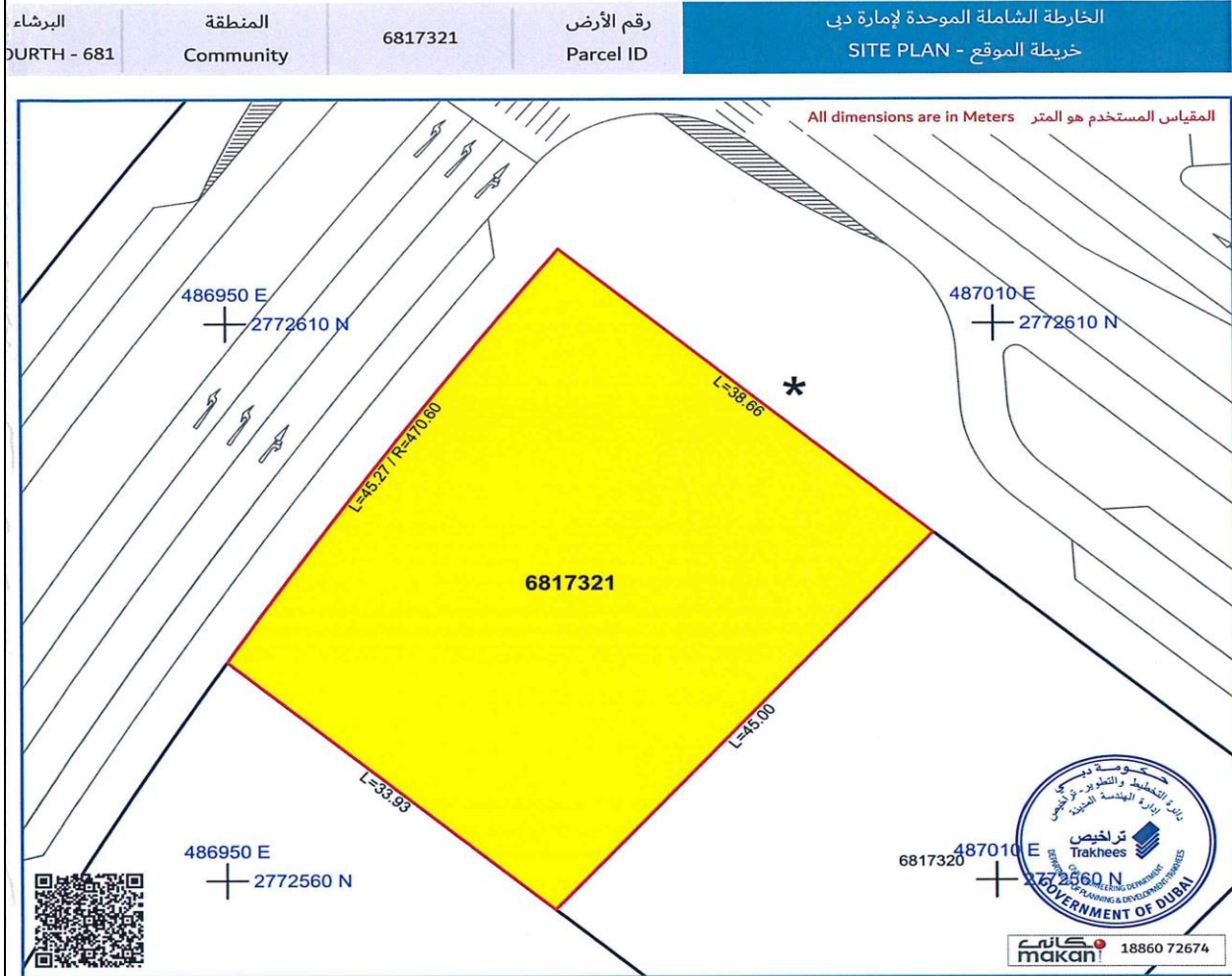
PLATE No: A.1.2

Plate 1.2 Site Location Plan

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Website: www.soilanalysis.hagalamir.com / Email: hasalab@hagalamir.com, hasa.lab1@gmail.com



Please scan QR code for online validation of Site Plan

يرجى مسح (رمز الاستجابة السريع) لعرض صلاحية الخريطة التخطيطية الإلكترونية

For Dimensions and Plot Coordinates Contact Trakhees Civil Engineering Division (Planning Section), All Units are in Meters. The Lessee/Possessor Will be Responsible For The Checking Or Relocation of Services Found Within The Plot. All Works To Comply With Trakhees Building Regulations & Design Guidelines, Approved DCR & All Related Services Authorities.

JVC15AMRM004	رقم المطور Developer Plot No.	NAKHEEL	ملاحظات Remarks	تاريخ الإصدار Issue Date
			المطور Developer	4-1-2023
	MDP DEVELOPMENT LIMITED		المالك Owner	حياة المالك Owner Title
	MDP DEVELOPMENT LIMITED		حائز POSSESSOR	ملك PRIVATE
1649.61 SQ. M.	المساحة الكلية Total Area	NTS	مقياس الرسم Scale	*
			مدخل الأرض Access Side	Valid for 2 years only from the date of issue
			الصلاحية Validity	



SITE: Building (4B + G + 10 + R)
JVC15AMRM004, Al Barsha South Fourth
Dubai
United Arab Emirates

TITLE: Site Plan

JOB REF : SIR2023-0018

DATE: 14-Mar-23

CLIENT: M/s. MDP Development Limited

PLATE No: A.2

Plate 2 Site Affection Plan

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Website: www.soilanalysis.hagalamir.com / Email: hasalab@hagalamir.com, hasa.lab1@gmail.com

GENERAL NOTES:
1. UNLESS OTHERWISE NOTED, THE REQUIRED DEPTH OF BOREHOLES IS IN METERS.
2. THE CONSULTANT'S RESPONSIBILITY IS TO PROVIDE THE BOREHOLE LOCATION PLAN ISSUED BY THE CLIENT AND TO VERIFY THE BOREHOLE LOCATION PLAN ISSUED BY THE CLIENT.
3. ALL COORDINATES AND DISTANCES ARE IN METERS.
4. THE BOREHOLE LOCATION PLAN MUST BE CARRY OUT THE WORK CAREFULLY TO AVOID DAMAGE OF EXISTING SERVICES (IF ANY).
5. THE SOIL CONSULTANT MUST ASCERTAIN THAT THE PROVIDED COORDINATES AND DISTANCES ARE CORRECT AND TO BE USED TO LOCATE THE BOREHOLE LOCATION PLAN.
6.1. THE BOREHOLE LOCATION PLAN MUST BE USED TO LOCATE THE BOREHOLE LOCATION PLAN.
6.2. LOCATE ANY LEFT POCKETS IN THE SUPPORTING SOIL.



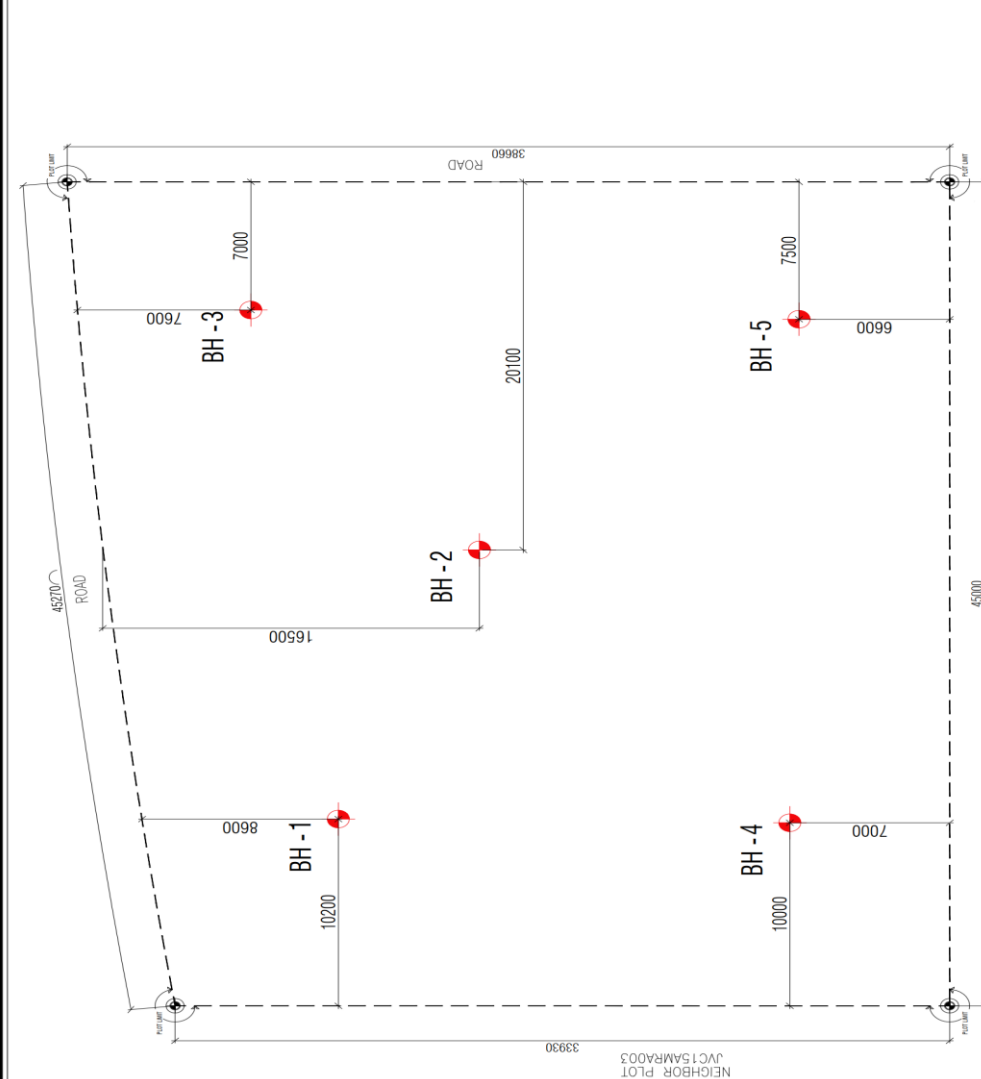
NO.	DATE	DESCRIPTION	BY	CHKD.
1.				
2.				
3.				
4.				
5.				
6.				
7.				
8.				
9.				
10.				
11.				
12.				

CLIENT: MDP DEVELOPMENT LIMITED

PROJECT:	RESIDENTIAL BUILDING (B+G+10+R)
DESIGNER:	MDP DEVELOPMENT LIMITED
CONTRACTOR:	MDP DEVELOPMENT LIMITED
CLIENT:	MDP DEVELOPMENT LIMITED
LOT NO.:	JVC15AMRM004

MAIN CONSULTANT:	FEDERAL ENGINEERING CONSULTANTS
PROJECT NO.:	DXB-2009
DESIGNED BY:	ALJAFRI
ARCH. ENG.:	SALEH
MEP:	SALEH
DRAWING TITLE:	BOREHOLE LOCATION

DRAWING NO.:	BH-00
DATE:	JAN 2023
CHECKED BY:	1:100
SCALE:	1:100
REV. NO.:	001



BOREHOLE DETAILS		
NO.	DISTANCE	DEPTH (METER)
BH-01	REFER TO LAYOUT	30.00
BH-02	REFER TO LAYOUT	40.00
BH-03	REFER TO LAYOUT	30.00
BH-04	REFER TO LAYOUT	30.00
BH-05	REFER TO LAYOUT	30.00

NEIGHBOR PLOT
JVC15AMRM006

1 BOREHOLE LOCATION
SCALE 1:100



JOB REF : SIR2023-0018

DATE: 14-Mar-23

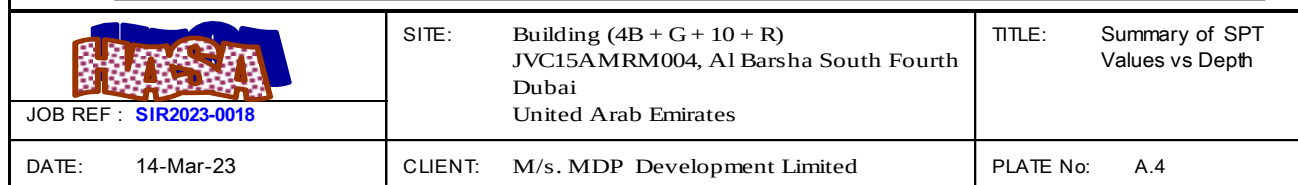
SITE: Building (4B + G + 10 + R)
JVC15AMRM004, Al Barsha South Fourth
Dubai
United Arab Emirates

CLIENT: M/s. MDP Development Limited

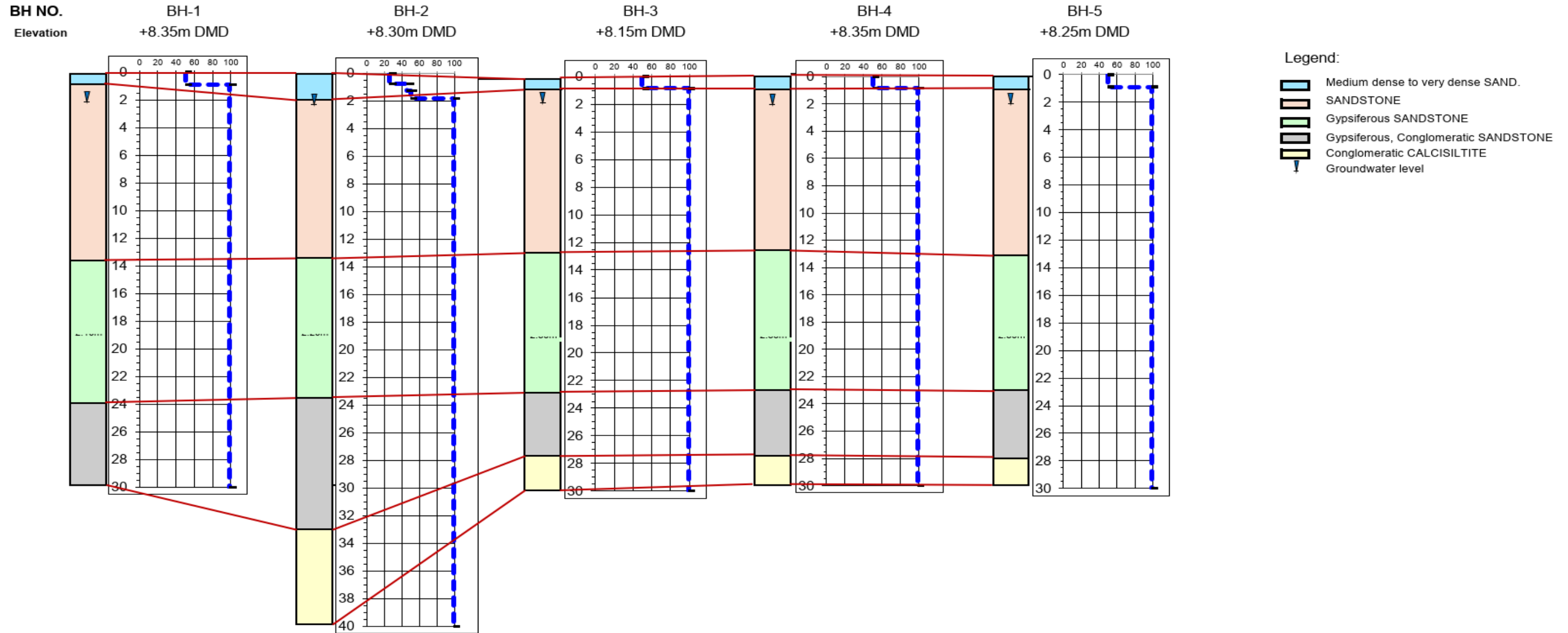
TITLE: Borehole Location Plan

PLATE No: A.3

Plate 3: Boreholes Location Plan.



Report Ref. No. SIR2023-0018 – Rev. No. 00



JOB REF : SIR2023-0018 DXB

DATE: 14-Mar-23

SITE: Building (4B + G + 10 + R)
JVC15AMRM004, Al Barsha South Fourth
Dubai
United Arab Emirates

CLIENT: M/s. MDP Development Limited

TITLE: Geotechnical Cross Section Profile

PLATE No: A.5

Plate 5 Geotechnical Cross Section Profile

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APPENDIX B (FIELD RESULTS)

BOREHOLE LOGS



PLATE 6 TO 10

BOREHOLE LOG

BOREHOLE NO. 01
SHEET NO. 1 of 3

Client				M/s MDP Development Limited.						Drilled By		Name	HS		
Project				Building (4B +G +D +R)								Sign			
Location				Plot No. JVC 15AM RM 004 (6817321), Al Barsha South Fourth, Dubai, U.A.E.						Logged By		Name	TY		
Ground Datum Level				+8.35m DMD.								Sign			
Date Drilling Started				27-02-2023						Checked By		Name	MG		
Date Drilling Completed				28-02-2023								Sign			
Drilling Method				Percussion from 0.0m ~ 0.89m & Rotary from 0.89m ~ 30.0m.						Coordinates		N			
Drilling Fluid Used				Water/ Bentonite								E			
Method	Casing Size & Depth (m)	Reduced Level (m)	Water Depth (m)	Depth (m)	Legend	DESCRIPTION	Sample		SPT	UCS, N/mm2	TCR (%)	SCR (%)	RQD (%)	STRUCTURAL DETAILS Bedding, Foliation, Lamination, Joints, Inclusions, Slip Planes, Spacing, Fabric, Seams, Discontinuities, Shear Zones	
						Method	Depth (m)								
CPB			▽ 2.10m	0.5	X X	Very dense, light brown to brown, silty, fine to medium sand with cemented & sandstone pieces.	D-1	GL	13,27,23* N=50/235mm	2.66				Very close to medium spaced horizontal fractures.	
0.89				X		S-1	0.5								
10						0.89		C-1			191	90	72		32
15						2.00		C-2				87	67		20
2.0						2.00									
2.5						3.50									
3.0						3.50		C-3				85	69		29
4.0						5.00									
5.0						5.00		C-4				87	73		37
6.0						6.50									
7.0						6.50		C-5				93	77		27
8.0						8.00									
9.0						8.00		C-6				99	77		53
9.50															
10.0						10.00									

Continued

WATER LEVEL (m)			REMARKS :
Date	Time	Water Level	
28-02-23	4:30 p.m.	2.10m	
Key			Note - SPT N value should be recorded for each 15cm run as well as the total of the last two. Vane test should include 3 readings : undisturbed, after 5 Minutes and after 15 minutes
TCR-Total Core Recovery			
SCR - Solid Core Recovery			
RQD-Rock Quality Designation			
SPT-Standard Penetration Test			
HSA - Hollow Stem Auger			
RCTB-RotaryCoring with T. Carbide Bit			
RCDB-Rotary Coring with Diamond Bit			
HWL -HQ Wire Line			
PLATE: 6.1			

Continued

1 Borehole was drilled up to 30.0m depth from existing ground level.

Report No.: SIR2023-0018

Plate 6.1 Borehole Log for Borehole No. 1

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BOREHOLE LOG

BOREHOLE NO. 01
SHEET NO. 2 of 3

Client				M/s MDP Development Limited.						Drilled By		Name	HS		
Project				Building (4B +G + 10 +R)								Sign			
Borehole Location				Plot No. JVC 15AM RM 004 (681732), Al Barsha South Fourth, Dubai, U.A.E.						Logged By		Name	TY		
Ground Datum Level				+8.35m DMD.								Sign			
Date Drilling Started				27-02-2023						Checked By		Name	MG		
Date Drilling Completed				28-02-2023								Sign			
Drilling Method				Percussion from 0.0m ~ 0.89m & Rotary from 0.89m ~ 30.0m.						Coordinates		N			
Drilling Fluid Used				Water/ Bentonite								E			
Method	Casing size & Depth (m)	Reduced Level (m)	Water Depth (m)	Depth (m)	Legend	DESCRIPTION	Sample		SPT	UCS, N/mm2	TCR (%)	SCR (%)	RQD (%)	STRUCTURAL DETAILS	
						Method	Depth (m)	Bedding, Foliation, Lamination, Joints, Inclusions, Slip Planes, Spacing, Fabric, Seams, Discontinuities, Shear Zones							
RCDB				10.0		Same as previous		10.00						Very close to medium spaced horizontal fractures.	
				11.0		Extremely weak to very weak, light brown to brown, very thinly to medium bedded, fine to medium grained, SANDSTONE.	C-7	11.00			90	79	35		
				12.0			C-8	11.00							
				13.0				12.50							
				13.5			C-9	12.50		2.55	93	77	57		
				14.0		Very weak, brown to light reddish brown, thinly to medium bedded, fine to medium grained, slightly to gypsiferous SANDSTONE.		14.00							
				15.0			C-10	14.00				88	80		49
				16.0				15.50							
				16.0			C-11	15.50				87	81		60
				17.0				17.00		2.97					
				18.0			C-12	17.00				88	81		61
				19.0				18.50							
				20.0			C-13	18.50				87	80		63

Continued

WATER LEVEL (m)			REMARKS :
Date	Time	Water Level	
28-02-23	4:30 p.m.	2.10m	
Key			
TCR-Total Core Recovery		D- Disturbed Sample	Note - SPT N value should be recorded for each 15cm run as well as the total of the last two . Vane test should include 3 readings : undisturbed, after 5 M inutes and after 15 minutes
SCR-Solid Core Recovery		UD - Undisturbed Sample	
RQD-Rock Quality Designation		BD - Bulk Disturbed Sample	
SPT-Standard Penetration Test		SFA - Solid Flight Auger	
HSA - Hollow Stem Auger		CPB - Cable Percussion Boring	
RCTB -RotaryCoring with T. Carbide Bit		RWB - Rotary Wash Boring	
RCDB-Rotary Coring with Diamond Bit			
HWL -HQ Wire Line			
PLATE: 6.2			Report No.: SIR2023-0018

Plate 6.2 Borehole Log for Borehole No. 1

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BOREHOLE LOG

BOREHOLE NO. 01
SHEET NO. 3 of 3

Client				M/s MDP Development Limited.						Drilled By		Name	HS	
Project				Building (4B +G +D +R)								Sign		
Borehole Location				Plot No. JVC 15AM RM 004 (6817321), Al Barsha South Fourth, Dubai, U.A.E.						Logged By		Name	TY	
Ground Datum Level				+8.35m DM D.								Sign		
Date Drilling Started				27-02-2023						Checked By		Name	MG	
Date Drilling Completed				28-02-2023								Sign		
Drilling Method				Percussion from 0.0m ~ 0.89m & Rotary from 0.89m ~ 30.0m.						Coordinates		N		
Drilling Fluid Used				Water/ Bentonite								E		
Method	Casing 0 m Size & Depth	Reduced Level (m)	Water Depth (m)	Depth (m)	Legend	DESCRIPTION	Sample		SPT	UCS, N/mm2	TCR (%)	SCR (%)	RQD (%)	STRUCTURAL DETAILS Bedding, Foliation, Lamination, Joints, Inclusions, Slip Planes, Spacing, Fabric, Seams, Discontinuities, Shear Zones
							Method	Depth (m)						
RCDB				20.0		Same as previous. Very weak, brown to light reddish brown, thinly to medium bedded, fine to medium grained, slightly to gypsiferous SANDSTONE.	C-14	20.0	2.88		88	82	66	Closely to medium spaced horizontal fractures.
				21.0		21.5								
				22.0		21.5								
				23.0		23.0								
				23.9		23.0								
				24.0		C-16	24.5							
				25.0		24.5								
				26.0		C-17	26.0							
				27.0		26.0								
				28.0		C-18	27.5							
				29.0		27.5								
150mm 29.5m				30.0			C-20	29.0	158		80	60	50	

WATER LEVEL (m)			REMARKS :	
Date	Time	Water Level	1 Borehole was drilled up to 30.0m depth from existing ground level.	
28-02-23	4:30 p.m.	2.10m		
Key			Note - SPT N value should be recorded for each 15cm run as well as the total of the last two. Vane test should include 3 readings : undisturbed, after 5 Minutes and after 15 minutes	
TCR - Total Core Recovery			R - Refusal	
SCR - Solid Core Recovery			Cw - Core waxed Sample	
RQD - Rock Quality Designation				
SPT - Standard Penetration Test				
HSA - Hollow Stem Auger				
RCTB - Rotary Coring with Tricone Bit				
RCDB - Rotary Coring with Diamond Bit				
HWL - HQ Wire Line				
PLATE: 6.3			Report No.: SIR2023-0018	

Plate 6.3 Borehole Log for Borehole No. 1

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BOREHOLE LOG

BOREHOLE NO. BH-02
SHEET NO. 1 of 4

Client		M/s MDP Development Limited.								Drilled By		Name		HS	
Project		Building (4B +G + 1D +R)										Sign			
Location		Plot No. JVC 15A MRM 004 (6817321), Al Barsha South Fourth, Dubai, U.A.E.								Logged By		Name		TY	
Ground Datum Level		+8.30m DMD.										Sign			
Date Drilling Started		22-02-2023								Checked By		Name		SHK	
Date Drilling Completed		23-02-2023										Sign			
Equipment Type		Percussion from 0.0m ~ 187m & Rotary from 1.87m ~ 40.0m.								Coordinates		N			
Drilling Fluid Used		Water/ Bentonite										E			
Method	Casing Size & Depth (m)	Reduced Level (m)	Water Depth (m)	Depth (m)	Legend	DESCRIPTION	Sample		SPT	UCS N/mm2	TCR (%)	SCR (%)	RQD (%)	STRUCTURAL DETAILS Bedding, Foliation, Lamination, Joints, Inclusions, Slip Planes, Spacing, Fabric, Seams, Discontinuities, Shear Zones	
							Method	Depth (m)							
CPB			2.10m	10	X X	Medium dense to dense, becoming very dense, light brown to brown, silty, fine to medium SAND with cemented pieces from depth 150m.	B-1	GL	5,10,16 N=26 14,18,28 N=46 17,30,20* N=50/225mm						
							S-2	0.5							
							S-3	10							
							S-4	15							
RCDB			2.10m	187	X	Very weak, light brown to brown, very thinly to medium bedded, fine to medium grained SANDSTONE.		187		126	88	53	18	Close to medium spaced, horizontal to subhorizontal fractures.	
				2.0											
				2.5			C-1								
				3.0				3.00							
								3.00							
				4.0			C-2								
				5.0				4.50							
								4.50							
				6.0			C-3								
								6.00							
								6.00							
				7.0			C-4								
8.0			7.50												
			7.50												
9.0		C-5													
			9.00												
			9.00												
10.0		C-6													

Continued

WATER LEVEL (m)			REMARKS : 1 Borehole was completed to 40.0m depth from existing ground level.
Date	Time	Water Level	
23-02-23	4:00 p.m.	2.10m	
Key			Note - SPT N value should be recorded for each 15cm run as well as the total of the last two. Vane test should include 3 readings : undisturbed, after 5 Minutes and after 15 minutes
TCR-Total Core Recovery			
SCR - Solid Core Recovery		D- Disturbed Sample	
RQD-Rock Quality Designation		UD - Undisturbed Sample	
SPT-Standard Penetration Test		BD - Bulk Disturbed Sample	
HSA - Hollow Stem Auger		SFA - Solid Flight Auger	
RCTB-RotaryCoring with T. Carbide		CPB - Cable Percussion Boring	
RCDB-Rotary Coring with Diamond		RWB- Rotary Wash Boring	
HWL -HQ Wire Line			
PLATE: 7.1			Report No.: SIR2023-0018

Plate 7.1 Borehole Log for Borehole No. 2

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BOREHOLE LOG

BOREHOLE NO. BH-02
SHEET NO. 2 of 4

Client				M/s MDP Development Limited.						Drilled By		Name	HS				
Project				Building (4B +G +10 +R)								Sign					
Borehole Location				Plot No. JVC 15AMRM 004 (681732), Al Barsha South Fourth, Dubai, U.A.E.						Logged By		Name	TY				
Ground Datum Level				+8.50m DMD.								Sign					
Date Drilling Started				22-02-2023						Checked By		Name	SHK				
Date Drilling Completed				23-02-2023								Sign					
Equipment Type				Percussion from 0.0m ~ 187m & Rotary from 1.87m ~ 40.0m.						Coordinates		N					
Drilling Fluid Used				Water/ Bentonite								E					
Method	Casing size & Depth (m)	Reduced Level (m)	Water Depth (m)		Legend	DESCRIPTION		Sample			SPT	UCS N/mm2	TCR (%)	SCR (%)	RQD (%)	STRUCTURAL DETAILS	
RCDB						(as previous sheet)										Bedding, Foliation, Lamination, Joints, Inclusions, Slip Planes, Spacing, Fabric, Seams, Discontinuities, Shear Zones	
				10.0		Very weak, light brown to brown, very thinly to medium bedded, fine to medium grained SANDSTONE.										Close to medium spaced, horizontal to sub-horizontal features.	
				11.0					10.50								
				12.0					12.00								
				12.0													
				13.0													
				13.5													
				14.0			Very weak, brown to light reddish brown, thinly to medium bedded, fine to medium grained, slightly to gypsiferous SANDSTONE.										
				15.0													
				16.0													
				17.0													
				18.0													
			19.0														
			20.0														

WATER LEVEL (m)

Date	Time	Water Level
23-02-23	4:00 p.m.	2.10m

1 Borehole was completed to 40.0m depth from existing ground level.

Key

TCR - Total Core Recovery
SCR - Solid Core Recovery
RQD - Rock Quality Designation
SPT - Standard Penetration Test
HSA - Hollow Stem Auger
RCTB - Rotary Coring with T. Carbide
RCDB - Rotary Coring with Diamond Bit
HWL - HQ Wire Line
D - Disturbed Sample
UD - Undisturbed Sample
BD - Bulk Disturbed Sample
SFA - Solid Flight Auger
CPB - Cable Percussion Boring
RWB - Rotary Wash Boring

R - Refusal
Cw - Core waxed Sample

Note - SPT N value should be recorded for each 15cm run as well as the total of the last two. Vane test should include 3 readings : undisturbed, after 5 Minutes and after 15 minutes

PLATE: 7.2

Report No.: SIR2023-0018

Plate 7.2 Borehole Log for Borehole No. 2

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BOREHOLE LOG

BOREHOLE NO. BH-02

SHEET NO. 3 of 4

Client		M/s MDP Development Limited.						Drilled By		Name	HS			
Project		Building (4B +G +10 +R)								Sign				
Borehole Location		Plot No. JVC 15AM RM 004 (6817321), Al Barsha South Fourth, Dubai, U.A.E.						Logged By		Name	TY			
Ground Datum Level		+8.30m DMD.								Sign				
Date Drilling Started		22-02-2023						Checked By		Name	SHK			
Date Drilling Completed		23-02-2023								Sign				
Equipment Type		Percussion from 0.0m ~ 187m & Rotary from 1.87m ~ 40.0m.						Coordinates		N				
Drilling Fluid Used		Water/ Bentonite								E				
Method	Casing 0 m Size & Depth	Reduced Level (m)	Water Depth (m)	Depth (m)	Legend	DESCRIPTION	Sample		SPT	UCS N/mm2	TCR (%)	SCR (%)	RQD (%)	STRUCTURAL DETAILS Bedding, Foliation, Lamination, Joints, Inclusions, Slip Planes, Spacing, Fabric, Seams, Discontinuities, Shear Zones
							Method	Depth (m)						
RCDB				20.0		Very weak, light brown to brown, very thinly to medium bedded, fine to medium grained, slightly to gypsiferous SANDSTONE.	C-13			2.00	93	84	62	Medium to wide, occasionally closed spaced horizontal fractures.
		21.00												
		21.00												
	C-14													
		22.50												
		22.0				22.50								
		23.0												
		23.3				22.50								
		24.0												
		24.0		Very weak to weak, light brown to brown, thinly to medium grained, slightly gypsiferous, conglomeratic SANDSTONE.	C-15		91	73			57			
		24.00												
		24.00												
	C-16													
		25.50												
		25.0				25.50								
		26.0												
	26.0				25.50									
	27.0													
	27.0				27.00									
	28.0													
	28.0				27.00									
	29.0													
	29.0				28.00									
	30.0													
	30.0				28.50									
	30.00													

WATER LEVEL (m)			REMARKS :
Date	Time	Water Level	
23-02-23	4:00 p.m.	2.10m	1 Borehole was completed to 40.0m depth from existing ground level.

Key TCR - Total Core Recovery SCR - Solid Core Recovery RQD - Rock Quality Designation SPT - Standard Penetration Test HSA - Hollow Stem Auger RCTB - Rotary Coring with Tricone B RCDB - Rotary Coring with Diamond Bit HWL - HQ Wire Line PLATE: 7.3	D - Disturbed Sample UD - Undisturbed Sample BD - Bulk Disturbed Sample SFA - Solid Flight Auger CPB - Cable Percussion Boring RWB - Rotary Wash Boring	R - Refusal Cw - Core waxed Sample	Note - SPT N value should be recorded for each 15cm run as well as the total of the last two. Vane test should include 3 readings : undisturbed, after 5 Minutes and after 15 minutes
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Report No.: SIR2023-0018

Plate 7.3 Borehole Log for Borehole No. 2

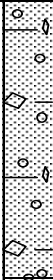
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BOREHOLE LOG

BOREHOLE NO. BH-02
SHEET NO. 4 of 4

Client					M/s MDP Development Limited.					Drilled By		Name	HS	
Project					Building (4B + G + 10 + R)							Sign		
Borehole Location					Plot No. JVC 15AM RM 004 (6817321), Al Barsha South Fourth, Dubai, U.A.E.					Logged By		Name	TY	
Ground Datum Level					+8.50m DMD.							Sign		
Date Drilling Started					22-02-2023					Checked By		Name	SHK	
Date Drilling Completed					23-02-2023							Sign		
Equipment Type					Percussion from 0.0m ~ 187m & Rotary from 1.87m ~ 40.0m.					Coordinates		N		
Drilling Fluid Used					Water/ Bentonite							E		
Method	Casing 0 m Size & Depth	Reduced Level (m)	Water Depth (m)	Depth (m)	Legend	DESCRIPTION	Sample		SPT	UCS N/mm2	TCR (%)	SCR (%)	RQD (%)	STRUCTURAL DETAILS
							Method	Depth (m)						
RCDB				30.0		(same as previous sheet) Very weak, light brown to brown, thinly to medium bedded, fine to medium grained, slightly gypsiferous, conglomeratic SANDSTONE.	C-19	30.00		2.51	91	80	60	Medium to widely spaced horizontal fractures.
				31.0		31.50								
				32.0		31.50								
				33.0		33.00								
				34.0		33.00								
				35.0		34.50								
				35.0		34.50								
				36.0		36.00								
				36.0		36.00								
				37.0		37.50								
37.0	37.50													
38.0	37.50													
38.0	37.50													
39.0	39.00													
39.0	39.00													
40.0	40.00													
40.0	40.00													
	150mm 39.5m					Weak, light brown to creamyish brown, thinly to medium bedded, conglomeratic CALCISILTITE.	C-21	34.50		10.35	83	73	47	
34.0	34.50													
35.0	34.50													
36.0	36.00													
36.0	36.00													
37.0	37.50													
37.0	37.50													
38.0	37.50													
38.0	37.50													
39.0	39.00													
39.0	39.00													
40.0	40.00													
40.0	40.00													
							C-22	36.00		6.36	83	73	43	
34.0	34.50													
35.0	34.50													
36.0	36.00													
36.0	36.00													
37.0	37.50													
37.0	37.50													
38.0	37.50													
38.0	37.50													
39.0	39.00													
39.0	39.00													
40.0	40.00													
40.0	40.00													
							C-23	37.50		1153	90	75	45	
34.0	34.50													
35.0	34.50													
36.0	36.00													
36.0	36.00													
37.0	37.50													
37.0	37.50													
38.0	37.50													
38.0	37.50													
39.0	39.00													
39.0	39.00													
40.0	40.00													
40.0	40.00													
							C-24	39.00						
34.0	34.50													
35.0	34.50													
36.0	36.00													
36.0	36.00													
37.0	37.50													
37.0	37.50													
38.0	37.50													
38.0	37.50													
39.0	39.00													
39.0	39.00													
40.0	40.00													
40.0	40.00													
							C-25	40.00						
34.0	34.50													
35.0	34.50													
36.0	36.00													
36.0	36.00													
37.0	37.50													
37.0	37.50													
38.0	37.50													
38.0	37.50													
39.0	39.00													
39.0	39.00													
40.0	40.00													
40.0	40.00													

WATER LEVEL (m)			REMARKS :
Date	Time	Water Level	
23-02-23	4:00 p.m.	2.10m	

Key TCR - Total Core Recovery SCR - Solid Core Recovery RQD - Rock Quality Designation SPT - Standard Penetration Test HSA - Hollow Stem Auger RCTB - Rotary Coring with Tricone Bit RCDB - Rotary Coring with Diamond Bit HWL - HQ Wire Line PLATE : 7.4	D - Disturbed Sample UD - Undisturbed Sample BD - Bulk Disturbed Sample SFA - Solid Flight Auger CPB - Cable Percussion Boring RWB - Rotary Wash Boring	R - Refusal Cw - Core waxed Sample Note - SPT N value should be recorded for each 15cm run as well as the total of the last two. Vane test should include 3 readings : undisturbed, after 5 Minutes and after 15 minutes
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Report No.: SIR2023-0018

Plate 7.4 Borehole Log for Borehole No. 2

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BOREHOLE LOG

BOREHOLE NO. 03
SHEET NO. 1 of 3

Client				M/s MDP Development Limited.						Drilled By		Name	HS				
Project				Building (4B +G +10 +R)								Sign					
Location				Plot No. JVC 15AMRM 004 (6817321), Al Barsha South Fourth, Dubai, U.A.E.						Logged By		Name	TY				
Ground Datum Level				+8.15m DMD.								Sign					
Date Drilling Started				28-02-2023						Checked By		Name	MG				
Date Drilling Completed				01-03-2023								Sign					
Drilling Method				Percussion from 0.0m ~ 0.86m & Rotary from 0.86m ~ 30.0m.						Coordinates		N					
Drilling Fluid Used				Water/ Bentonite								E					
Method	Casing Size & Depth (m)	Reduced Level (m)	Water Depth (m)	Depth (m)	Legend	DESCRIPTION	Sample		SPT	UCS, N/mm2	TCR (%)	SCR (%)	RQD (%)	STRUCTURAL DETAILS Bedding, Foliation, Lamination, Joints, Inclusions, Slip Planes, Spacing, Fabric, Seams, Discontinuities, Shear Zones			
						Method	Depth (m)										
CPB	RCDB		▽ 2.00m	0.5	X	Very dense, light brown to brown, silty, fine to medium sand with cemented & sandstone pieces.	D-1	GL	17,33,17* N=50/210mm								
				0.86	X		S-1	0.5									
				10					0.86	166							
				15					C-1								
				2.0					2.00								
				2.5					2.00								
				3.0						C-2				85	73	50	
				4.0							3.50						
											3.50						
				5.0						C-3				83	75	23	
											5.00						
											5.00						
				6.0						C-4				90	74	27	
											6.50						
											6.50						
				7.0						C-5				82	73	30	
											8.00						
											8.00						
				9.0						C-6				80	67	24	
								9.50									
								10.00									

Continued

WATER LEVEL (m)			REMARKS : 1 Borehole was drilled up to 30.0m depth from existing ground level.
Date	Time	Water Level	
01-03-23	4:32 p.m.	2.00m	
Key			
TCR-Total Core Recovery			V - Vane R- Refusal Cw - Core waxed Sample
SCR - Solid Core Recovery			
RQD-Rock Quality Designation			
SPT-Standard Penetration Test			
HSA - Hollow Stem Auger			
RCTB -RotaryCoring with T. Carbide Bit			
RCDB -Rotary Coring with Diamond Bit			Note - SPT N value should be recorded for each 15cm run as well as the total of the last two. Vane test should include 3 readings : undisturbed, after 5 Minutes and after 15 minutes
HWL -HQ Wire Line			
PLATE: 8.1			
			Report No.: SIR2023-0018

Report No.: SIR2023-0018

Plate 8.1 Borehole Log for Borehole No. 3

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BOREHOLE LOG

BOREHOLE NO. 03

SHEET NO. 2 of 3

Client					M/s MDP Development Limited.					Drilled By		Name	HS			
Project					Building (4B +G +10 +R)							Sign				
Borehole Location					Plot No. JVC 15AM RM 004 (681732), Al Barsha South Fourth, Dubai, U.A.E.					Logged By		Name	TY			
Ground Datum Level					+8.15m DMD.							Sign				
Date Drilling Started					28-02-2023					Checked By		Name	MG			
Date Drilling Completed					01-03-2023							Sign				
Drilling Method					Percussion from 0.0m ~ 0.86m & Rotary from 0.86m ~ 30.0m.					Co ordinates		N				
Drilling Fluid Used					Water/ Bentonite							E				
Method	Casing size & Depth (m)	Reduced Level (m)	Water Depth (m)	Depth (m)	Legend	DESCRIPTION	Sample		SPT	UCS, N/mm2	TCR (%)	SCR (%)	RQD (%)	STRUCTURAL DETAILS Bedding, Foliation, Lamination, Joints, Inclusions, Slip Planes, Spacing, Fabric, Seams, Discontinuities, Shear Zones		
							Method	Depth (m)								
RCDB				10.0		Same as previous Extremely weak to very weak, light brown to brown, very thinly to medium bedded, fine to medium grained, SANDSTONE.	C-7	10.00		0.98	87	69	33	Close to medium spaced horizontal fractures.		
				11.0			11.00									
				12.0			11.00	C-8				73	53		13	
				12.75			12.50				12.50					
				13.0			Extremely weak to cery weak, brown to light r reddish brown, thinly to medium bedded, fine to medium grained, slightly gypsiferous to gypsiferous SANDSTONE.	C-9				85	70		37	
				14.0				14.00				14.00				
				15.0				15.50			C-10		87		77	57
				16.0				16.50				16.50				
				17.0				17.00			C-11		91		80	53
				18.0				18.50				18.50				
				19.0				19.50			C-12		96		86	62
				20.0				20.00			C-13		92		82	66

Continued

WATER LEVEL (m)			REMARKS :
Date	Time	Water Level	
01-03-23	4:32 p.m.	2.00m	1 Borehole was drilled up to 30.0m depth from existing ground level.
Key TCR-Total Core Recovery SCR-Solid Core Recovery RQD-Rock Quality Designation SPT-Standard Penetration Test HSA - Hollow Stem Auger RCTB-Rotary Coring with T. Carbide Bit RCDB-Rotary Coring with Diamond Bit HWL -HQ Wire Line PLATE: 8.2			D- Disturbed Sample UD - Undisturbed Sample BD - Bulk Disturbed Sample SFA - Solid Flight Auger CPB - Cable Percussion Boring R - Refusal Cw - Core waxed Sample Note - SPT N value should be recorded for each 15cm run as well as the total of the last two. Vane test should include 3 readings : undisturbed, after 5 Minutes and after 15 minutes
Report No.: SIR2023-0018			

Plate 8.2 Borehole Log for Borehole No. 3

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BOREHOLE LOG

BOREHOLE NO. 03
SHEET NO. 3 of 3

Client	M/s MDP Development Limited.	Drilled By	Name	HS
Project	Building (4B +G +1D +R)		Sign	
Borehole Location	Plot No. JVC15AMRM 004 (681732), Al Barsha South Fourth, Dubai, U.A.E.	Logged By	Name	TY
Ground Datum Level	+8.15m DMD.		Sign	
Date Drilling Started	28-02-2023	Checked By	Name	MG
Date Drilling Completed	01-03-2023		Sign	
Drilling Method	Percussion from 0.0m ~ 0.86m & Rotary from 0.86m ~ 30.0m.	Coordinates	N	
Drilling Fluid Used	Water/ Bentonite		E	

Method	Casing 0 m Size & Depth	Reduced Level (m)	Water Depth (m)	Depth (m)	Legend	DESCRIPTION	Sample		SPT	UCS, N/mm2	TCR (%)	SCR (%)	RQD (%)	STRUCTURAL DETAILS Bedding, Foliation, Lamination, Joints, Inclusions, Slip Planes, Spacing, Fabric, Seams, Discontinuities, Shear Zones	
							Method	Depth (m)							
RCDB	150mm 29.50m			20.0		Same as previous. Very weak, brown to light reddish brown, thinly to medium bedded, fine to medium grained, slightly to gypsiferous SANDSTONE.	C-14	20.0		2.99	90	83	65	Closely to medium spaced horizontal fractures.	
				21.0		215	215								
				22.0				C-15				100	87		73
				22.8											
				23.0							23.0				
				24.0				C-16				87	77		59
				24.5							24.5				
				25.0				C-17				87	75		59
				26.0							26.0				
				27.0				C-18				82	70		55
				27.6							27.5				
				28.0				C-19				77	67		59
				29.0							29.0				
				29.5				C-20				81	70		58
				30.0							30.0				

E.O.B.

WATER LEVEL (m)			REMARKS :
Date	Time	Water Level	1 Borehole was drilled up to 30.0m depth from existing ground level.
01-03-23	4:32 p.m.	2.00m	

Key TCR-Total Core Recovery SCR-Solid Core Recovery RQD-Rock Quality Designation SPT-Standard Penetration Test HSA - Hollow Stem Auger RCTB-Rotary Coring with Tricone Bit RCDB-Rotary Coring with Diamond Bit HWL -HQ Wire Line PLATE : 8.3	D- Disturbed Sample UD - Undisturbed Sample BD - Bulk Disturbed Sample SFA - Solid Flight Auger CPB - Cable Percussion Boring RWB - Rotary Wash Boring	R - Refusal Cw - Core waxed Sample	Note - SPT N value should be recorded for each 15cm run as well as the total of the last two. Vane test should include 3 readings : undisturbed, after 5 Minutes and after 15 minutes
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Report No.: SIR2023-0018

Plate 8.3 Borehole Log for Borehole No. 3

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BOREHOLE LOG

BOREHOLE NO. 04
SHEET NO. 1 of 3

Client					M/s MDP Development Limited.					Drilled By		Name	HS	
Project					Building (4B +G + 10 +R)							Sign		
Location					Plot No. JVC 15AM RM 004 (681732), Al Barsha South Fourth, Dubai, U.A.E.					Logged By		Name	TY	
Ground Datum Level					+8.35m DMD.							Sign		
Date Drilling Started					27-02-2023					Checked By		Name	MG	
Date Drilling Completed					28-02-2023							Sign		
Drilling Method					Percussion from 0.0m ~ 0.85m & Rotary from 0.85m ~ 30.0m.					Coordinates		N		
Drilling Fluid Used					Water/ Bentonite							E		
Method	Casing Size & Depth (m)	Reduced Level (m)	Water Depth (m)	Depth (m)	Legend	DESCRIPTION	Sample		SPT	UCS, N/mm2	TCR (%)	SCR (%)	RQD (%)	STRUCTURAL DETAILS Bedding, Foliation, Lamination, Joints, Inclusions, Slip Planes, Spacing, Fabric, Seams, Discontinuities, Shear Zones
CPB				0.5	X	Very dense, light brown to brown, silty, fine to medium sand with cemented & sandstone pieces.	D-1	GL						
				0.85	X		S-1	0.5	2130,20* N=50/200mm					
				10										
				15		Extremely weak to very weak, light brown to brown, very thinly to medium bedded, fine to medium grained, SANDSTONE.	C-1	0.85			87	70	00	Very close to medium spaced horizontal fractures.
				2.0				2.00						
				2.5			C-2	2.00		0.91				
				3.0							83	67	30	
				3.5				3.50						
				4.0			C-3	3.50			82	67	30	
				5.0				5.00						
				6.0			C-4	5.00			87	73	33	
				6.5				6.50		2.40				
				7.0				6.50						
				8.0			C-5				93	77	39	
				8.0				8.00						
				9.0			C-6	8.00			87	73	13	
				9.5				9.50						
				10.0				10.00		189				

Continued

WATER LEVEL (m)			REMARKS :	
Date	Time	Water Level		
28-02-23	4:30 p.m.	2.15m		
Key TCR-Total Core Recovery SCR - Solid Core Recovery RQD-Rock Quality Designation SPT-Standard Penetration Test HSA - Hollow Stem Auger RCTB-RotaryCoring with T. Carbide Bit RCDB-Rotary Coring with Diamond Bit HWL -HQ Wire Line			V - Vane R - Refusal Cw - Core waxed Sample	Note - SPT N value should be recorded for each 15cm run as well as the total of the last two. Vane test should include 3 readings : undisturbed, after 5 Minutes and after 15 minutes
P L A T E : 9.1			Report No.: SIR2023-0018	

Plate 9.1 Borehole Log for Borehole No. 4

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BOREHOLE LOG

BOREHOLE NO. 04

SHEET NO. 2 of 3

Client				M/s MDP Development Limited.						Drilled By		Name	HS		
Project				Building (4B +G +10 +R)								Sign			
Borehole Location				Plot No. JVC 15A MRM 004 (681732), Al Barsha South Fourth, Dubai, U.A.E.						Logged By		Name	TY		
Ground Datum Level				+8.35m DMD.								Sign			
Date Drilling Started				27-02-2023						Checked By		Name	MG		
Date Drilling Completed				28-02-2023								Sign			
Drilling Method				Percussion from 0.0m ~ 0.85m & Rotary from 0.85m ~ 30.0m.						Coordinates		N			
Drilling Fluid Used				Water/ Bentonite								E			
Method	Casing size & Depth (m)	Reduced Level (m)	Water Depth (m)	Depth (m)	Legend	DESCRIPTION	Sample		SPT	UCS, N/mm2	TCR (%)	SCR (%)	RQD (%)	STRUCTURAL DETAILS Bedding, Foliation, Lamination, Joints, Inclusions, Slip Planes, Spacing, Fabric, Seams, Discontinuities, Shear Zones	
							Method	Depth (m)							
RCDB				10.0		Same as previous Extremely weak to very weak, light brown to brown, very thinly to medium bedded, fine to medium grained, SANDSTONE.	C-7	10.00		4.10	85	67	20	Very close to medium spaced horizontal fractures.	
						11.0		11.00							
				12.0			C-8				93	79	34		
				12.5				12.50							
				13.0			Very weak, brown to light reddish brown, thinly to medium bedded, fine to medium grained, slightly to gypsiferous SANDSTONE.	C-9				87	79		57
				14.0							14.00				
								14.00							
				15.0				C-10				88	78		25
											15.50				
								15.50							
				16.0				C-11				94	80		59
											17.00				
								17.00							
18.0			C-12		87	73	56								
				18.50											
			18.50												
19.0			C-13		91	80	62								
				20.00											
			20.00												

Continued

WATER LEVEL (m)			REMARKS :
Date	Time	Water Level	
28-02-23	4:30 p.m.	2.15m	
Key			
TCR-Total Core Recovery			D- Disturbed Sample
SCR-Solid Core Recovery			UD - Undisturbed Sample
RQD-Rock Quality Designation			BD - Bulk Disturbed Sample
SPT-Standard Penetration Test			SFA - Solid Flight Auger
HSA - Hollow Stem Auger			CPB - Cable Percussion Boring
RCTB -RotaryCoring with T. Carbide Bit			RWB - Rotary Wash Boring
RCDB -Rotary Coring with Diamond Bit			
HWL -HQ Wire Line			
P L A T E : 9.2			Report No.: SIR2023-0018

Plate 9.2 Borehole Log for Borehole No. 4

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BOREHOLE LOG

BOREHOLE NO. 04
SHEET NO. 3 of 3

Client					M/s MDP Development Limited.					Drilled By		Name		HS			
Project					Building (4B +G +1D +R)							Sign					
Borehole Location					Plot No. JVC15AMRM 004 (681732), Al Barsha South Fourth, Dubai, U.A.E.					Logged By		Name		TY			
Ground Datum Level					+8.35m DM D.							Sign					
Date Drilling Started					27-02-2023					Checked By		Name		MG			
Date Drilling Completed					28-02-2023							Sign					
Drilling Method					Percussion from 0.0m ~ 0.85m & Rotary from 0.85m ~ 30.0m.					Coordinates		N					
Drilling Fluid Used					Water/ Bentonite							E					
Method	Casing 0 m Size & Depth	Reduced Level (m)	Water Depth (m)	Depth (m)	Legend	DESCRIPTION	Sample		SPT	UCS, N/mm2	TCR (%)	SCR (%)	RQD (%)	STRUCTURAL DETAILS Bedding, Foliation, Lamination, Joints, Inclusions, Slip Planes, Spacing, Fabric, Seams, Discontinuities, Shear Zones			
							Method	Depth (m)									
RCDB	150mm 29.5m			20.0		Same as previous. Weak, light brown to brown, thinly to medium bedded, fine to medium grained, gypsiferous SANDSTONE.	C-14	20.0		3.55		93	81	61	Closely to medium spaced horizontal fractures.		
				21.0		215 215											
				22.0		C-15	23.0	23.0									
				23.0			23.0										
				24.0		C-16	24.5 24.5	87								73	63
				25.0			C-17										
				26.0		C-18	27.5 27.5	80								73	56
				27.0			C-19										
				28.0		C-20	30.0	55								45	35
				29.0			30.0										

WATER LEVEL (m)			REMARKS :
Date	Time	Water Level	1 Borehole was drilled up to 30.0m depth from existing ground level.
28-02-23	4:30 p.m.	2.15m	

Key TCR-Total Core Recovery SCR-Solid Core Recovery RQD-Rock Quality Designation SPT-Standard Penetration Test HSA - Hollow Stem Auger RCTB-Rotary Coring with Tricone Bit RCDB-Rotary Coring with Diamond Bit HWL -HQ Wire Line PLATE : 9.3	D- Disturbed Sample UD - Undisturbed Sample BD - Bulk Disturbed Sample SFA - Solid Flight Auger CPB - Cable Percussion Boring RWB - Rotary Wash Boring	R- Refusal Cw - Core waxed Sample Note - SPT N value should be recorded for each 15cm run as well as the total of the last two. Vane test should include 3 readings : undisturbed, after 5 Minutes and after 15 minutes
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Report No.: SIR2023-0018

Plate 9.3 Borehole Log for Borehole No. 4

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BOREHOLE LOG

BOREHOLE NO. 05
SHEET NO. 1 of 3

Client				M/s MDP Development Limited.						Drilled By		Name	HS		
Project				Building (4B + G + 10 + R)								Sign			
Location				Plot No. JVC 15AMRM 004 (6817321), Al Barsha South Fourth, Dubai, U.A.E.						Logged By		Name	TY		
Ground Datum Level				+8.25m DMD.								Sign			
Date Drilling Started				28-02-2023						Checked By		Name	MG		
Date Drilling Completed				01-03-2023								Sign			
Drilling Method				Percussion from 0.0m ~ 0.90m & Rotary from 0.89m ~ 30.0m.						Co ordinates		N			
Drilling Fluid Used				Water/ Bentonite								E			
Method	Casing Size & Depth (m)	Reduced Level (m)	Water Depth (m)	Depth (m)	Legend	DESCRIPTION	Sample		SPT	UCS, N/mm2	TCR (%)	SCR (%)	RQD (%)	STRUCTURAL DETAILS Bedding, Foliation, Lamination, Joints, Inclusions, Slip Planes, Spacing, Fabric, Seams, Discontinuities, Shear Zones	
							Method	Depth (m)							
CPB			▽ 2.10m	0.5		Very dense, light brown to brown, silty, fine to medium sand with cemented & sandstone pieces.	D-1	GL	15,26,24* N=50/250mm	162					
				0.90			S-1	0.5							
RCDB				1.0		1.5	Extremely weak to very weak, light brown to brown, very thinly to medium bedded, fine to medium grained, SANDSTONE.	C-1	0.90			91	68		18
				2.0		2.00									
				2.5		2.00		C-2	3.50			92	80		42
				3.0		3.50									
				3.5		3.50		C-3	5.00			90	77		37
				4.0		5.00									
				5.0		5.00		C-4	6.50			95	79		21
				6.0		6.50									
7.0	6.50	C-5	8.00	85	69	34									
8.0	8.00														
9.0	8.00	C-6	9.50	93	79	29									
10.0	9.50														
								10.00							

Continued

WATER LEVEL (m)			REMARKS : 1 Borehole was drilled up to 30.0m depth from existing ground level.
Date	Time	Water Level	
01-03-23	4:00 p.m.	2.10m	
Key TCR-Total Core Recovery SCR - Solid Core Recovery RQD-Rock Quality Designation SPT-Standard Penetration Test HSA - Hollow Stem Auger RCTB -RotaryCoring with T. Carbide Bit RCDB-RotaryCoring with Diamond Bit HWL -HQ Wire Line			V - Vane R - Refusal Cw- Core waxed Sample <

Report No.: SIR2023-0018

Plate 10.1 Borehole Log for Borehole No. 5

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BOREHOLE LOG

BOREHOLE NO. 05

SHEET NO. 2 of 3

Client				M/s MDP Development Limited.						Drilled By		Name		HS	
Project				Building (4B + G + 10 + R)								Sign			
Borehole Location				Plot No. JVC 15AMRM 004 (681732), Al Barsha South Fourth, Dubai, U.A.E.						Logged By		Name		TY	
Ground Datum Level				+8.25m DMD.								Sign			
Date Drilling Started				28-02-2023						Checked By		Name		MG	
Date Drilling Completed				01-03-2023								Sign			
Drilling Method				Percussion from 0.0m – 0.90m & Rotary from 0.89m – 30.0m.						Co ordinates		N			
Drilling Fluid Used				Water/ Bentonite								E			
RCDB	Method	Casing size & Depth (m)	Reduced Level (m)	Water Depth (m)	Depth (m)	Legend	DESCRIPTION	Sample		SPT	UCS, N/mm2	TCR (%)	SCR (%)	RQD (%)	STRUCTURAL DETAILS Bedding, Foliation, Lamination, Joints, Inclusions, Slip Planes, Spacing, Fabric, Seams, Discontinuities, Shear Zones
								Method	Depth (m)						
					10.0		Same as previous Extremely weak to very weak, light brown to brown, very thinly to medium bedded, fine to medium grained, SANDSTONE.	C-7	10.00						
				11.0	11.00										
				12.0	11.00										
				13.0	12.50										
				13.2		Very weak, brown to light reddish brown, thinly to medium bedded, fine to medium grained, slightly to gypsiferous SANDSTONE.	C-9	12.50							
				14.0				14.00							
				14.0				14.00							
				15.0				15.50							
				16.0			C-10	15.50							
				16.0				15.50							
				17.0				17.00							
			17.0	17.00											
			18.0			C-11	17.00								
			18.0				17.00								
			19.0				18.50								
			19.0				18.50								
			20.0			C-12	18.50								
			20.0				18.50								
											195	85	75	57	Very close to medium spaced horizontal fractures

Continued

WATER LEVEL (m)			REMARKS :	
Date	Time	Water Level	1 Borehole was drilled up to 30.0m depth from existing ground level.	
01-03-23	4:00 p.m.	2.10m		
Key				
TCR-Total Core Recovery			D- Disturbed Sample	R- Refusal
SCR-Solid Core Recovery			UD - Undisturbed Sample	Cw - Core waxed Sample
RQD-Rock Quality Designation			BD - Bulk Disturbed Sample	Note - SPT N value should be recorded for each 15cm run as well as the total of the last two. Vane test should include 3 readings : undisturbed, after 5 M inutes and after 15 minutes
SPT-Standard Penetration Test			SFA - Solid Flight Auger	
HSA - Hollow Stem Auger			CPB - Cable Percussion Boring	
RCTB-Rotary Coring with T. Carbide Bit			RWB - Rotary Wash Boring	
RCDB-Rotary Coring with Diamond Bit				
HWL -HQ Wire Line				
P L A T E : 10.2			Report No.: SIR2023-0018	

Report No.: SIR2023-0018

Plate 10.2 Borehole Log for Borehole No. 5

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BOREHOLE LOG

BOREHOLE NO. 05
SHEET NO. 3 of 3

Client		M/s M DP Development Limited.						Drilled By		Name		HS			
Project		Building (4B +G +10 +R)								Sign					
Borehole Location		Plot No. JVC15AMRM 004 (6817321), Al Barsha South Fourth, Dubai, U.A.E.						Logged By		Name		TY			
Ground Datum Level		+8.25m DMD.								Sign					
Date Drilling Started		28-02-2023						Checked By		Name		MG			
Date Drilling Completed		01-03-2023								Sign					
Drilling Method		Percussion from 0.0m ~ 0.90m & Rotary from 0.89m ~ 30.0m.						Coordinates		N					
Drilling Fluid Used		Water/ Bentonite								E					
Method	Casing O m Size & Depth	Reduced Level (m)	Water Depth (m)	Depth (m)	Legend	DESCRIPTION	Sample		SPT	UCS, N/mm ²	TCR (%)	SCR (%)	RQD (%)	STRUCTURAL DETAILS Bedding, Foliation, Lamination, Joints, Inclusions, Slip Planes, Spacing, Fabric, Seams, Discontinuities, Shear Zones	
							Method	Depth (m)							
RCDB	150mm 29.50m			20.0		Same as previous. Weak, light brown to brown, thinly to medium bedded, fine to medium grained, gypsiferous SANDSTONE.	C-14	20.0		4.15		87	77	59	Closely to medium spaced horizontal fractures.
				21.0		215									
				22.0		C-15	215	95				87	73		
				23.0			23.0								
				24.0		C-16	23.0	80				73	57		
				24.5			24.5								
				25.0		C-17	24.5	73				57	27		
				26.0			26.0								
				27.0		C-18	26.0	80				73	59		
				27.5			27.5								
28.0	C-19	27.5	87	75	57										
29.0		29.0													
30.0	C-20	29.0	75	58	48										
		30.0	6.05												

WATER LEVEL (m)			REMARKS :
Date	Time	Water Level	1 Borehole was drilled up to 30.0m depth from existing ground level.
01-03-23	4:00 p.m.	2.10m	

Key TCR - Total Core Recovery SCR - Solid Core Recovery RQD - Rock Quality Designation SPT - Standard Penetration Test HSA - Hollow Stem Auger RCTB - Rotary Coring with Tricone Bit RCDB - Rotary Coring with Diamond Bit HWL - HQ Wire Line PLATE: 10.3	D - Disturbed Sample UD - Undisturbed Sample BD - Bulk Disturbed Sample SFA - Solid Flight Auger CPB - Cable Percussion Boring RWB - Rotary Wash Boring	R - Refusal Cw - Core waxed Sample Note - SPT N value should be recorded for each 15cm run as well as the total of the last two. Vane test should include 3 readings : undisturbed, after 5 Minutes and after 15 minutes
--	--	---

Report No.: SIR2023-0018

Plate 10.3 Borehole Log for Borehole No. 5

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APPENDIX C (LABORATORY TEST RESULTS)

PARTICLE SIZE DISTRIBUTION

UNCONFINED COMPRESSIVE STRENGTH

CHEMICAL TEST RESULTS

PLATE 11 TO 12

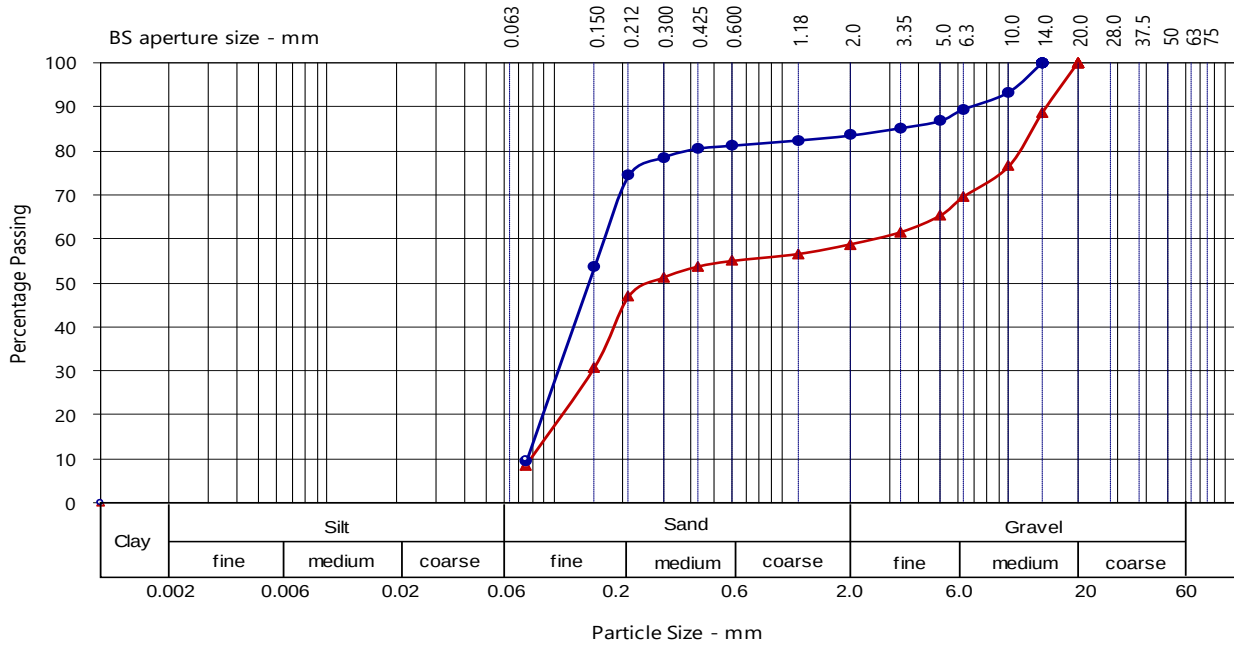
PLATE 13 TO 31

PLATE 32

PARTICLE SIZE DISTRIBUTION

Date of test started : 05th March 2023
Date of test finished : 08th March 2023
Report Ref. No. : SIR2023-0018
Plot No. – Location : JVC15AMRM004 – Al Barsha South Fourth
Client : M/s. MDP Development Limited.

Test method : BS 1377 Part 2: 1990: AMD 9027: 1996: Cl. 9.2 (wet sieving)
Sample prep. method : BS: 1377 Part 2: 1990: Cl 9.2.3.
Tested by : OU
Checked by : MG



Borehole No.	Sample #	Depth, m	Soil Description	Gravel, %	Sand, %	Silt & Clay, %
BH-01	S-02	0.50 - 0.89	Silty/Clayey, Very Gravelly, SAND	41	50	9
BH-02	S-02	0.50 - 0.95	Silty/Clayey, Gravelly, SAND	19	71	10

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PLATENo : 6

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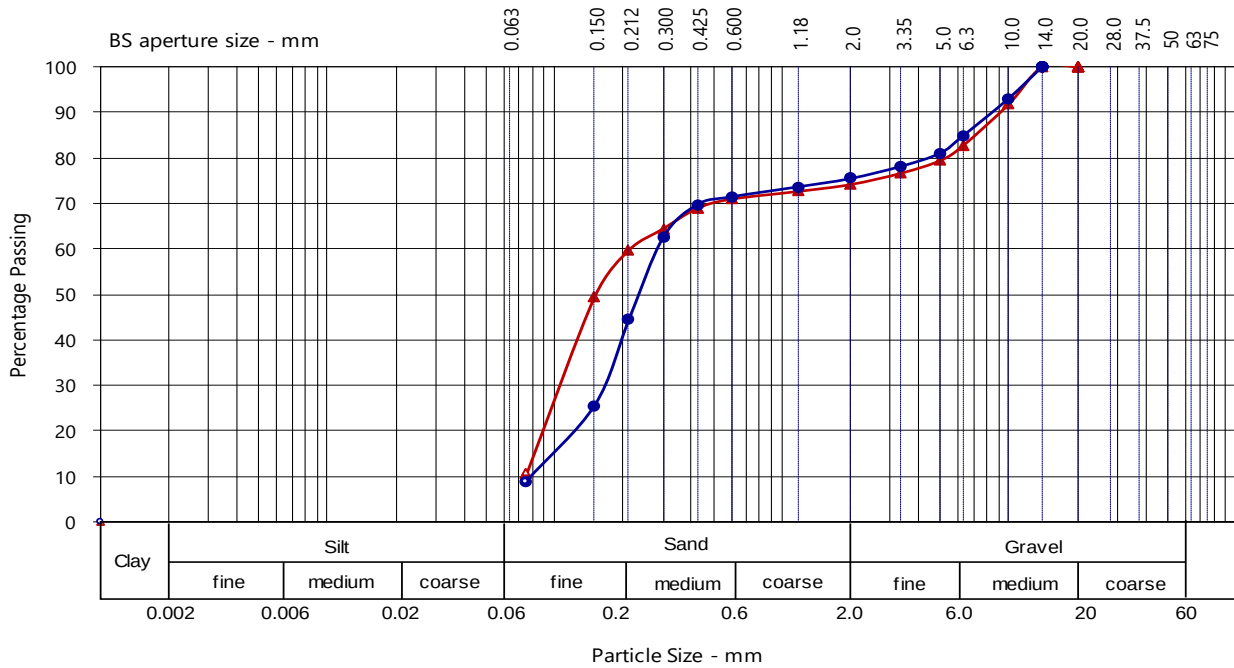
Plate 11: Particle Size Distribution for Borehole Nos. 1 & 2

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PARTICLE SIZE DISTRIBUTION

Date of test started : 05th March 2023
Date of test finished : 08th March 2023
Report Ref. No. : SIR2023-0018
Plot No. – Location : JVC15AMRM004 – Al Barsha South Fourth
Client : M/s. MDP Development Limited.

Test method : BS 1377 Part 2: 1990: AMD 9027: 1996: Cl. 9.2 (wet sieving)
Sample prep. method : BS: 1377 Part 2: 1990: Cl 9.2.3.
Tested by : OU
Checked by : MG



Borehole No.	Sample #	Depth, m	Soil Description	Gravel, %	Sand, %	Silt & Clay, %
BH-04	S-02	0.50 - 0.85	Silty/Clayey, Very Gravelly, SAND	26	63	11
BH-02	S-02	0.50 - 0.90	Silty/Clayey, Very Gravelly, SAND	28	63	9

Authorized Signatory

PLATENo : 6

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Plate 12: Particle Size Distribution for Borehole Nos. 4 & 5

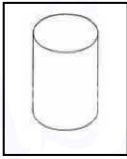
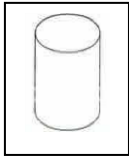
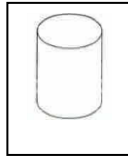
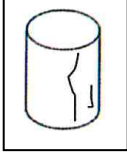
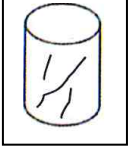
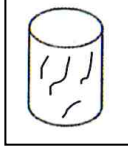
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UNCONFINED COMPRESSIVE STRENGTH TEST OF INTACT ROCK CORE REPORT

Date of sample received : 05th March 2023
Date of test started : 05th March 2023
Date of test finished : 08th March 2023
Plot No. – Location : JVC15AMRM004 – Al Barsha South Fourth.
Client : M/s. MDP Development Limited.

Report Ref. No. : SIR2023-0018
Test method : ASTM D7012-14
Sample prep. method : ASTM D4543-19
Tested by : SC.

Source / Borehole No.	BH-1	BH-1	BH-1
Sample No.	1	2	3
Depth, m	0.89 to 1.09	5.00 to 5.20	8.95 to 9.15

Straightness, S ₁ (Limits = ≤0.50mm)	<0.44	<0.45	<0.42
Flatness of ends, FP2 (Limits ≤25µm)	38	40	35
Perpendicularity, P2 (Limits = Δ/L ≤0.0043)	<0.0028	<0.0031	<0.0030
Mean length, mm	165.0	139.0	140.0
Mean diameter, mm	72.00	75.00	74.00
Length to diameter ratio (L/D)	2.29	1.85*	1.89*
Moisture condition	as received	as received	as received
Moisture content, %	9.4	16.60	22.9
Visual description / Lithology	SANDSTONE	SANDSTONE	SANDSTONE
Location of apparent weakness plane before test			
Failure pattern			

Sample temperature at test, °C	23.0	23	23
Rate of loading, mm/min	0.50	0.50	0.50
Time Taken for Failure (Min)	02:24	02:06	01:50
Unconfined compressive strength, MPa	2.66	1.91	0.59

*Notes: Length to diameter ratio <2.0 to 2.50

Method of sample preparation: ASTM D4543-19 Procedures /Straightness-S1, End Flatness FP2, Perpendicularity-P2

Test Method Variation: None

Remarks: Best effort achieved in End Flatness preparation

Test result relates only to the item tested.

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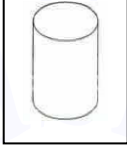
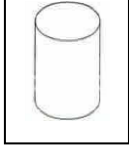
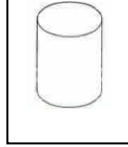
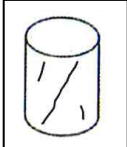
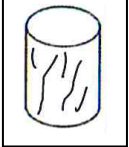
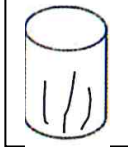
Plate 13: Unconfined Compressive Strength for Borehole No. 1

UNCONFINED COMPRESSIVE STRENGTH TEST OF INTACT ROCK CORE REPORT

Date of sample received : 05th March 2023
Date of test started : 05th March 2023
Date of test finished : 08th March 2023
Plot No. – Location : JVC15AMRM004 – Al Barsha South Fourth.
Client : M/s. MDP Development Limited.

Report Ref. No. : SIR2023-0018
Test method : ASTM D7012-14
Sample prep. method : ASTM D4543-19
Tested by : SC.

Source / Borehole No.	BH-1	BH-1	BH-1
Sample No.	4	5	6
Depth, m	13.00 to 13.20	16.80 to 17.00	20.00 to 20.20

Straightness, S ₁ (Limits = ≤0.50mm)	<0.48	<0.45	<0.46
Flatness of ends, FP2 (Limits ≤25μm)	36	38	35
Perpendicularity, P2 (Limits = Δ/L ≤0.0043)	<0.0034	<0.0039	<0.0036
Mean length, mm	184.00	186.00	190.00
Mean diameter, mm	75.00	76.00	76.00
Length to diameter ratio (L/D)	2.45	2.45	2.50
Moisture condition	as received	as received	as received
Moisture content, %	26.3	23.9	16.1
Visual description / Lithology	SANDSTONE	Gypsiferous SANDSTONE	Gypsiferous SANDSTONE
Location of apparent weakness plane before test			
Failure pattern			

Sample temperature at test, °C	23.0	23	23
Rate of loading, mm/min	0.50	0.50	0.50
Time Taken for Failure (Min)	02:16	02:34	02:21
Unconfined compressive strength, MPa	2.55	2.97	2.88

*Notes: Length to diameter ratio <2.0 to 2.50

Method of sample preparation: ASTM D4543-19 Procedures /Straightness-S1, End Flatness FP2, Perpendicularity-P2

Test Method Variation: None

Remarks: Best effort achieved in End Flatness preparation

Test result relates only to the item tested.

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Plate 14: Unconfined Compressive Strength for Borehole No. 1

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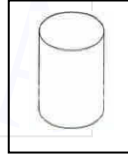
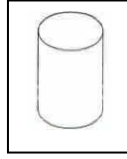
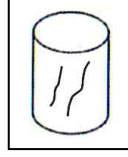
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UNCONFINED COMPRESSIVE STRENGTH TEST OF INTACT ROCK CORE REPORT

Date of sample received : 05th March 2023 Report Ref. No. : SIR2023-0018
Date of test started : 05th March 2023 Test method : ASTM D7012-14
Date of test finished : 08th March 2023 Sample prep. method : ASTM D4543-19
Plot No. – Location : JVC15AMRM004 – Al Barsha South Fourth. Tested by : SC.
Client : M/s. MDP Development Limited.

Source / Borehole No.	BH-1	BH-1
Sample No.	7	8
Depth, m	27.30 to 27.50	29.80 to 30.00

Straightness, S ₁ (Limits $\leq 0.50\text{mm}$)	<0.44	<0.46
Flatness of ends, FP2 (Limits $\leq 25\mu\text{m}$)	36	40
Perpendicularity, P2 (Limits $= \Delta/L \leq 0.0043$)	<0.0030	<0.0028
Mean length, mm	181.00	138.00
Mean diameter, mm	76.00	75.00
Length to diameter ratio (L/D)	2.38	1.84*
Moisture condition	as received	as received
Moisture content, %	24.6	16.1
Visual description / Lithology	Gypsiferous SANDSTONE	Conglomeratic SANDSTONE
Location of apparent weakness plane before test		
Failure pattern		

Sample temperature at test, °C	23.0	23.0
Rate of loading, mm/min	0.5	0.5
Time Taken for Failure (Min)	02.50	02:05
Unconfined compressive strength, MPa	3.21	1.58

*Notes: Length to diameter ratio <2.0 to 2.50

Method of sample preparation: ASTM D4543-19 Procedures /Straightness-S1, End Flatness FP2, Perpendicularity-P2

Test Method Variation: None

Remarks: Best effort achieved in End Flatness preparation

Test result relates only to the item tested.

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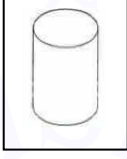
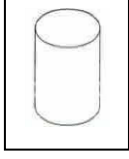
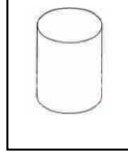
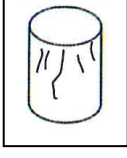
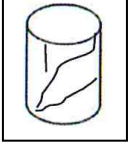
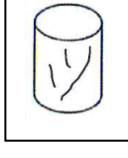
Plate 15: Unconfined Compressive Strength for Borehole No. 1

UNCONFINED COMPRESSIVE STRENGTH TEST OF INTACT ROCK CORE REPORT

Date of sample received : 05th March 2023
Date of test started : 05th March 2023
Date of test finished : 08th March 2023
Plot No. – Location : JVC15AMRM004 – Al Barsha South Fourth.
Client : M/s. MDP Development Limited.

Report Ref. No. : SIR2023-0018
Test method : ASTM D7012-14
Sample prep. method : ASTM D4543-19
Tested by : SC.

Source / Borehole No.	BH-2	BH-2	BH-2
Sample No.	1	2	3
Depth, m	2.60 to 2.80	6.20 to 6.40	11.80 to 12.00

Straightness, S ₁ (Limits $\leq 0.50\text{mm}$)	<0.38	<0.41	<0.42
Flatness of ends, FP2 (Limits $\leq 25\mu\text{m}$)	40	38	42
Perpendicularity, P2 (Limits $= \Delta/L \leq 0.0043$)	<0.0025	<0.0028	<0.0034
Mean length, mm	125.00	102.00	130.00
Mean diameter, mm	74.00	75.00	75.00
Length to diameter ratio (L/D)	1.69*	1.36*	1.73*
Moisture condition	as received	as received	as received
Moisture content, %	15.40	16.5	16.2
Visual description / Lithology	SANDSTONE	SANDSTONE	SANDSTONE
Location of apparent weakness plane before test			
Failure pattern			

Sample temperature at test, °C	23.0	23	23
Rate of loading, mm/min	0.50	0.50	0.50
Time Taken for Failure (Min)	01:44	01:55	02:36
Unconfined compressive strength, MPa	1.26	1.35	2.63

*Notes: Length to diameter ratio <2.0 to 2.50

Method of sample preparation: ASTM D4543-19 Procedures /Straightness-S1, End Flatness FP2, Perpendicularity-P2

Test Method Variation: None

Remarks: Best effort achieved in End Flatness preparation

Test result relates only to the item tested.

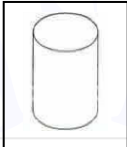
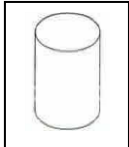
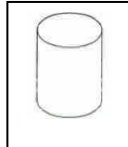
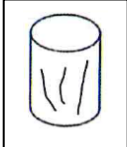
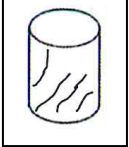
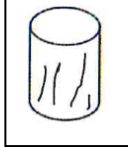
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Plate 16: Unconfined Compressive Strength for Borehole No. 2

UNCONFINED COMPRESSIVE STRENGTH TEST OF INTACT ROCK CORE REPORT

Date of sample received : 05th March 2023 Report Ref. No. : SIR2023-0018
Date of test started : 05th March 2023 Test method : ASTM D7012-14
Date of test finished : 08th March 2023 Sample prep. method : ASTM D4543-19
Plot No. – Location : JVC15AMRM004 – Al Barsha South Fourth. Tested by : SC.
Client : M/s. MDP Development Limited.

Source / Borehole No.	BH-2	BH-2	BH-2
Sample No.	4	5	6
Depth, m	15.00 to 15.20	9.30 to 19.50	22.50 to 22.70

Straightness, S ₁ (Limits = ≤0.50mm)	<0.45	<0.36	<0.42
Flatness of ends, FP2 (Limits ≤25μm)	40	42	36
Perpendicularity, P2 (Limits = Δ/L ≤0.0043)	<0.0028	<0.0031	<0.0029
Mean length, mm	162.00	162.00	179.00
Mean diameter, mm	75.00	75.00	76.00
Length to diameter ratio (L/D)	2.16	2.16	2.36
Moisture condition	as received	as received	as received
Moisture content, %	24.6	29.3	26.3
Visual description / Lithology	Gypsiferous SANDSTONE	Gypsiferous SANDSTONE	Gypsiferous SANDSTONE
Location of apparent weakness plane before test			
Failure pattern			

Sample temperature at test, °C	23.0	23	23
Rate of loading, mm/min	0.50	0.50	0.50
Time Taken for Failure (Min)	02:06	01.40	02.18
Unconfined compressive strength, MPa	1.56	1.24	2.00

*Notes: Length to diameter ratio <2.0 to 2.50

Method of sample preparation: ASTM D4543-19 Procedures /Straightness-S1, End Flatness FP2, Perpendicularity-P2

Test Method Variation: None

Remarks: Best effort achieved in End Flatness preparation

Test result relates only to the item tested.

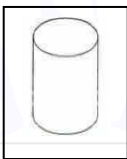
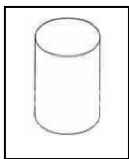
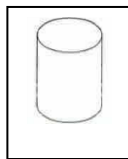
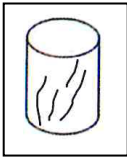
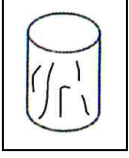
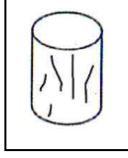
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Plate 17: Unconfined Compressive Strength for Borehole No. 2

UNCONFINED COMPRESSIVE STRENGTH TEST OF INTACT ROCK CORE REPORT

Date of sample received : 05th March 2023 Report Ref. No. : SIR2023-0018
Date of test started : 05th March 2023 Test method : ASTM D7012-14
Date of test finished : 08th March 2023 Sample prep. method : ASTM D4543-19
Plot No. – Location : JVC15AMRM004 – Al Barsha South Fourth. Tested by : SC.
Client : M/s. MDP Development Limited.

Source / Borehole No.	BH-2	BH-2	BH-2
Sample No.	7	8	9
Depth, m	27.00 to 27.20	31.30 to 31.50	34.35 to 34.50

Straightness, S ₁ (Limits = ≤0.50mm)	<0.45	<0.50	<0.46
Flatness of ends, FP2 (Limits ≤25μm)	44	45	42
Perpendicularity, P2 (Limits = Δ/L ≤0.0043)	<0.0036	<0.0038	<0.0035
Mean length, mm	186.00	181.00	122.00
Mean diameter, mm	75.00	76.00	75.00
Length to diameter ratio (L/D)	2.48	2.38	1.63*
Moisture condition	as received	as received	as received
Moisture content, %	18.2	26.8	4.40
Visual description / Lithology	Conglomeratic SANDSTONE	Conglomeratic SANDSTONE	CALCISILTITE
Location of apparent weakness plane before test			
Failure pattern			

Sample temperature at test, °C	23.0	23	23
Rate of loading, mm/min	0.50	0.50	0.50
Time Taken for Failure (Min)	03:42	02.32	03.48
Unconfined compressive strength, MPa	5.52	2.51	10.35

*Notes: Length to diameter ratio <2.0 to 2.50

Method of sample preparation: ASTM D4543-19 Procedures /Straightness-S1, End Flatness FP2, Perpendicularity-P2

Test Method Variation: None

Remarks: Best effort achieved in End Flatness preparation

Test result relates only to the item tested.

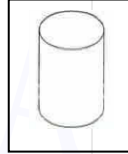
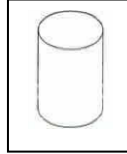
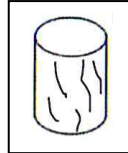
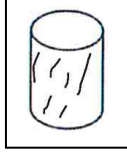
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Plate 18: Unconfined Compressive Strength for Borehole No. 2

UNCONFINED COMPRESSIVE STRENGTH TEST OF INTACT ROCK CORE REPORT

Date of sample received : 05th March 2023 Report Ref. No. : SIR2023-0018
Date of test started : 05th March 2023 Test method : ASTM D7012-14
Date of test finished : 08th March 2023 Sample prep. method : ASTM D4543-19
Plot No. – Location : JVC15AMRM004 – Al Barsha South Fourth. Tested by : SC.
Client : M/s. MDP Development Limited.

Source / Borehole No.	BH-2	BH-2
Sample No.	10	11
Depth, m	37.30 to 37.50	39.40 to 39.60

Straightness, S_1 (Limits $\leq 0.50\text{mm}$)	<0.47	<0.44
Flatness of ends, FP2 (Limits $\leq 25\mu\text{m}$)	44	41
Perpendicularity, P2 (Limits $= \Delta/L \leq 0.0043$)	<0.0035	<0.0040
Mean length, mm	109.00	181.00
Mean diameter, mm	74.00	75.00
Length to diameter ratio (L/D)	1.47*	2.41
Moisture condition	as received	as received
Moisture content, %	6.4	4.2
Visual description / Lithology	CALCISILTITE	CALCISILTITE
Location of apparent weakness plane before test		
Failure pattern		

Sample temperature at test, °C	23.0	23.0
Rate of loading, mm/min	0.5	0.5
Time Taken for Failure (Min)	03.26	04:16
Unconfined compressive strength, MPa	6.36	11.53

*Notes: Length to diameter ratio <2.0 to 2.50

Method of sample preparation: ASTM D4543-19 Procedures /Straightness-S1, End Flatness FP2, Perpendicularity-P2
Test Method Variation: None
Remarks: Best effort achieved in End Flatness preparation

Test result relates only to the item tested.

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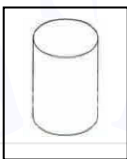
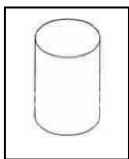
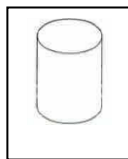
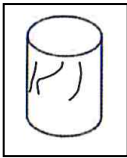
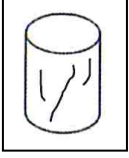
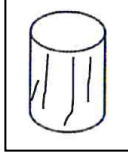
Plate 19: Unconfined Compressive Strength for Borehole No. 2

UNCONFINED COMPRESSIVE STRENGTH TEST OF INTACT ROCK CORE REPORT

Date of sample received : 05th March 2023
Date of test started : 05th March 2023
Date of test finished : 07th March 2023
Plot No. – Location : JVC15AMRM004 – Al Barsha South Fourth.
Client : M/s. MDP Development Limited.

Report Ref. No. : SIR2023-0018
Test method : ASTM D7012-14
Sample prep. method : ASTM D4543-19
Tested by : SC.

Source / Borehole No.	BH-3	BH-3	BH-3
Sample No.	1	2	3
Depth, m	12.00 to 12.50	6.30 to 6.50	14.10 to 14.30

Straightness, S ₁ (Limits = ≤0.50mm)	<0.46	<0.44	<0.4850
Flatness of ends, FP2 (Limits ≤25μm)	42	40	38
Perpendicularity, P2 (Limits = Δ/L ≤0.0043)	<0.0031	<0.0034	<0.0036
Mean length, mm	174.00	115.0	172.00
Mean diameter, mm	74.00	75.00	75.00
Length to diameter ratio (L/D)	2.35	1.53*	2.28
Moisture condition	as received	as received	as received
Moisture content, %	13.3	15.60	29.2
Visual description / Lithology	SANDSTONE	SANDSTONE	Gypsiferous SANDSTONE
Location of apparent weakness plane before test			
Failure pattern			

Sample temperature at test, °C	22.0	22	22
Rate of loading, mm/min	0.50	0.50	0.50
Time Taken for Failure (Min)	02:05	02:24	01:42
Unconfined compressive strength, MPa	1.66	1.82	0.98

*Notes: Length to diameter ratio <2.0 to 2.50

Method of sample preparation: ASTM D4543-19 Procedures /Straightness-S1, End Flatness FP2, Perpendicularity-P2

Test Method Variation: None

Remarks: Best effort achieved in End Flatness preparation

Test result relates only to the item tested.

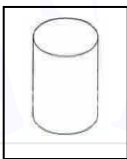
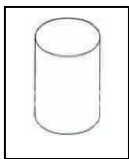
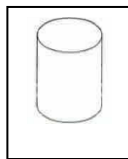
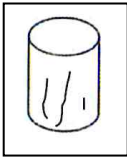
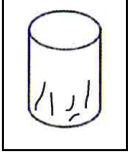
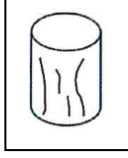
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Plate 20: Unconfined Compressive Strength for Borehole No. 3

UNCONFINED COMPRESSIVE STRENGTH TEST OF INTACT ROCK CORE REPORT

Date of sample received : 05th March 2023 Report Ref. No. : SIR2023-0018
Date of test started : 05th March 2023 Test method : ASTM D7012-14
Date of test finished : 07th March 2023 Sample prep. method : ASTM D4543-19
Plot No. – Location : JVC15AMRM004 – Al Barsha South Fourth. Tested by : SC.
Client : M/s. MDP Development Limited.

Source / Borehole No.	BH-3	BH-3	BH-3
Sample No.	4	5	6
Depth, m	18.50 to 18.70	23.00 to 23.20	29.00 to 29.20

Straightness, S ₁ (Limits = ≤0.50mm)	<0.40	<0.44	<0.46
Flatness of ends, FP2 (Limits ≤25μm)	36	38	35
Perpendicularity, P2 (Limits = Δ/L ≤0.0043)	<0.0026	<0.0028	<0.0034
Mean length, mm	162.00	167.00	175.00
Mean diameter, mm	75.00	75.00	75.00
Length to diameter ratio (L/D)	2.16	2.23	2.33
Moisture condition	as received	as received	as received
Moisture content, %	24.5	20.7	8.3
Visual description / Lithology	Gypsiferous SANDSTONE	Conglomeratic SANDSTONE	Conglomeratic CALCISILTITE
Location of apparent weakness plane before test			
Failure pattern			

Sample temperature at test, °C	22.0	22	22
Rate of loading, mm/min	0.50	0.50	0.50
Time Taken for Failure (Min)	02:35	02:48	03:52
Unconfined compressive strength, MPa	2.20	2.99	7.16

*Notes: Length to diameter ratio <2.0 to 2.50

Method of sample preparation: ASTM D4543-19 Procedures /Straightness-S1, End Flatness FP2, Perpendicularity-P2

Test Method Variation: None

Remarks: Best effort achieved in End Flatness preparation

Test result relates only to the item tested.

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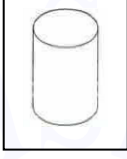
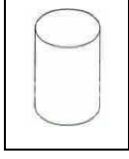
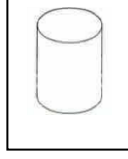
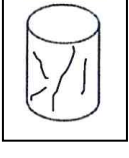
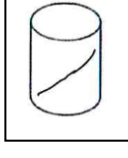
Plate 21: Unconfined Compressive Strength for Borehole No. 3

UNCONFINED COMPRESSIVE STRENGTH TEST OF INTACT ROCK CORE REPORT

Date of sample received : 05th March 2023
Date of test started : 05th March 2023
Date of test finished : 09th March 2023
Plot No. – Location : JVC15AMRM004 – Al Barsha South Fourth.
Client : M/s. MDP Development Limited.

Report Ref. No. : SIR2023-0018
Test method : ASTM D7012-14
Sample prep. method : ASTM D4543-19
Tested by : SC.

Source / Borehole No.	BH-4	BH-4	BH-4
Sample No.	1	2	3
Depth, m	2.10 to 2.30	6.30 to 6.50	9.50 to 9.70

Straightness, S ₁ (Limits $\leq 0.50\text{mm}$)	<0.46	<0.45	<0.42
Flatness of ends, FP2 (Limits $\leq 25\mu\text{m}$)	36	35	40
Perpendicularity, P2 (Limits $= \Delta/L \leq 0.0043$)	<0.0026	<0.0028	<0.0031
Mean length, mm	143.00	135.00	139.00
Mean diameter, mm	74.00	75.00	75.00
Length to diameter ratio (L/D)	1.93*	1.80*	1.85*
Moisture condition	as received	as received	as received
Moisture content, %	11.30	14.3	12.1
Visual description / Lithology	SANDSTONE	SANDSTONE	SANDSTONE
Location of apparent weakness plane before test			
Failure pattern			

Sample temperature at test, °C	22.0	22	22
Rate of loading, mm/min	0.50	0.50	0.50
Time Taken for Failure (Min)	01:44	02:12	02:12
Unconfined compressive strength, MPa	0.91	2.40	1.89

*Notes: Length to diameter ratio <2.0 to 2.50

Method of sample preparation: ASTM D4543-19 Procedures /Straightness-S1, End Flatness FP2, Perpendicularity-P2

Test Method Variation: None

Remarks: Best effort achieved in End Flatness preparation

Test result relates only to the item tested.

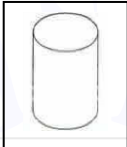
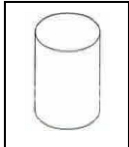
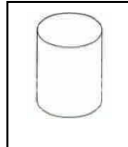
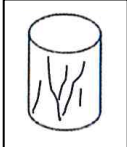
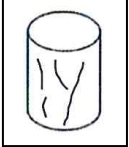
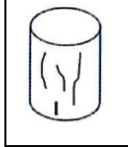
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Plate 22: Unconfined Compressive Strength for Borehole No. 4

UNCONFINED COMPRESSIVE STRENGTH TEST OF INTACT ROCK CORE REPORT

Date of sample received : 05th March 2023 Report Ref. No. : SIR2023-0018
Date of test started : 05th March 2023 Test method : ASTM D7012-14
Date of test finished : 09th March 2023 Sample prep. method : ASTM D4543-19
Plot No. – Location : JVC15AMRM004 – Al Barsha South Fourth. Tested by : SC.
Client : M/s. MDP Development Limited.

Source / Borehole No.	BH-4	BH-4	BH-4
Sample No.	4	5	6
Depth, m	15.30 to 15.70	18.50 to 18.70	23.00 to 23.20

Straightness, S ₁ (Limits = ≤0.50mm)	<0.45	<0.40	<0.41
Flatness of ends, FP2 (Limits ≤25μm)	44	38	36
Perpendicularity, P2 (Limits = Δ/L ≤0.0043)	<0.0034	<0.0029	<0.0028
Mean length, mm	184.00	182.00	155.00
Mean diameter, mm	75.00	76.00	75.00
Length to diameter ratio (L/D)	2.59	2.39	2.07
Moisture condition	as received	as received	as received
Moisture content, %	23.2	13.5	22.2
Visual description / Lithology	Gypsiferous SANDSTONE	Gypsiferous SANDSTONE	Conglomeratic SANDSTONE
Location of apparent weakness plane before test			
Failure pattern			

Sample temperature at test, °C	22.0	22	22
Rate of loading, mm/min	0.50	0.50	0.50
Time Taken for Failure (Min)	02:46	02:55	02.28
Unconfined compressive strength, MPa	4.10	4.85	3.55

*Notes: Length to diameter ratio <2.0 to 2.50

Method of sample preparation: ASTM D4543-19 Procedures /Straightness-S1, End Flatness FP2, Perpendicularity-P2

Test Method Variation: None

Remarks: Best effort achieved in End Flatness preparation

Test result relates only to the item tested.

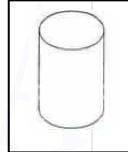
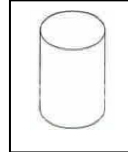
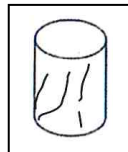
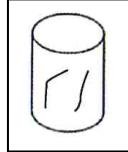
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Plate 23: Unconfined Compressive Strength for Borehole No. 4

UNCONFINED COMPRESSIVE STRENGTH TEST OF INTACT ROCK CORE REPORT

Date of sample received : 05th March 2023 Report Ref. No. : SIR2023-0018
Date of test started : 05th March 2023 Test method : ASTM D7012-14
Date of test finished : 09th March 2023 Sample prep. method : ASTM D4543-19
Plot No. – Location : JVC15AMRM004 – Al Barsha South Fourth. Tested by : SC.
Client : M/s. MDP Development Limited.

Source / Borehole No.	BH-4	BH-4
Sample No.	7	8
Depth, m	27.30 to 27.50	29.80 to 30.00

Straightness, S_1 (Limits $\leq 0.50\text{mm}$)	<0.42	<0.43
Flatness of ends, FP2 (Limits $\leq 25\mu\text{m}$)	31	32
Perpendicularity, P2 (Limits $= \Delta/L \leq 0.0043$)	<0.0028	<0.0034
Mean length, mm	158.00	110.00
Mean diameter, mm	75.00	75.00
Length to diameter ratio (L/D)	2.11	1.47*
Moisture condition	as received	as received
Moisture content, %	19.1	13.9
Visual description / Lithology	Conglomeratic SANDSTONE	CALCISILTITE
Location of apparent weakness plane before test		
Failure pattern		

Sample temperature at test, °C	22.0	22.0
Rate of loading, mm/min	0.5	0.5
Time Taken for Failure (Min)	02.52	02:20
Unconfined compressive strength, MPa	4.80	3.14

*Notes: Length to diameter ratio <2.0 to 2.50

Method of sample preparation: ASTM D4543-19 Procedures /Straightness-S1, End Flatness FP2, Perpendicularity-P2

Test Method Variation: None

Remarks: Best effort achieved in End Flatness preparation

Test result relates only to the item tested.

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Plate 24: Unconfined Compressive Strength for Borehole No. 4

UAE – Dubai – Al Quoz Ind. Third – Warehouse No. 22D.
Tel 04 – 341 6086 Fax 04 – 341 6096 - P.O. Box 510.

الإمارات العربية المتحدة – دبي – الفوز الصناعية الثالثة – شيرة رقم 22
تليفون 04 – 341 6086 فاكس 04 – 341 6096 – ص.ب. 510

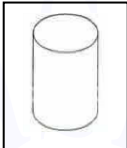
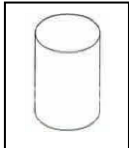
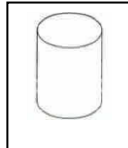
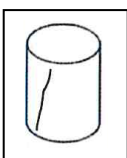
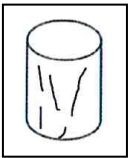
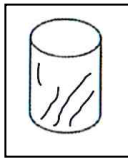
Website: www.soilanalysis.hagalamir.com / Email: hasalab@hagalamir.com, hasa.lab1@gmail.com

UNCONFINED COMPRESSIVE STRENGTH TEST OF INTACT ROCK CORE REPORT

Date of sample received : 05th March 2023
Date of test started : 05th March 2023
Date of test finished : 11th March 2023
Plot No. – Location : JVC15AMRM004 – Al Barsha South Fourth.
Client : M/s. MDP Development Limited.

Report Ref. No. : SIR2023-0018
Test method : ASTM D7012-14
Sample prep. method : ASTM D4543-19
Tested by : SC.

Source / Borehole No.	BH-5	BH-5	BH-5
Sample No.	1	2	3
Depth, m	2.00 to 2.20	7.10 to 7.30	12.30 to 12.50

Straightness, S ₁ (Limits = ≤0.50mm)	<0.48	<0.44	<0.45
Flatness of ends, FP2 (Limits ≤25μm)	45	40	41
Perpendicularity, P2 (Limits = Δ/L ≤0.0043)	<0.0034	<0.0028	<0.0026
Mean length, mm	161.00	108.00	143.00
Mean diameter, mm	74.00	74.00	75.00
Length to diameter ratio (L/D)	2.18	1.46*	1.91*
Moisture condition	as received	as received	as received
Moisture content, %	12.5	18.7	24.9
Visual description / Lithology	SANDSTONE	SANDSTONE	SANDSTONE
Location of apparent weakness plane before test			
Failure pattern			

Sample temperature at test, °C	22.0	22	22
Rate of loading, mm/min	0.50	0.50	0.50
Time Taken for Failure (Min)	02:05	02:03	01:56
Unconfined compressive strength, MPa	1.62	1.60	1.45

*Notes: Length to diameter ratio <2.0 to 2.50

Method of sample preparation: ASTM D4543-19 Procedures /Straightness-S1, End Flatness FP2, Perpendicularity-P2

Test Method Variation: None

Remarks: Best effort achieved in End Flatness preparation

Test result relates only to the item tested.

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Plate 25: Unconfined Compressive Strength for Borehole No. 5

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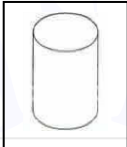
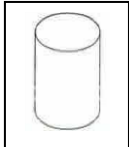
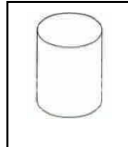
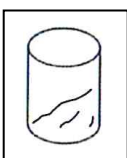
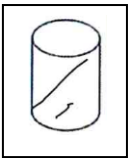
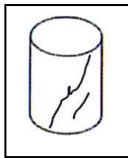
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UNCONFINED COMPRESSIVE STRENGTH TEST OF INTACT ROCK CORE REPORT

Date of sample received : 05th March 2023
Date of test started : 05th March 2023
Date of test finished : 11th March 2023
Plot No. – Location : JVC15AMRM004 – Al Barsha South Fourth.
Client : M/s. MDP Development Limited.

Report Ref. No. : SIR2023-0018
Test method : ASTM D7012-14
Sample prep. method : ASTM D4543-19
Tested by : SC.

Source / Borehole No.	BH-5	BH-5	BH-5
Sample No.	4	5	6
Depth, m	15.50 to 15.70	18.30 to 18.50	22.80 to 23.00

Straightness, S ₁ (Limits = ≤0.50mm)	<0.44	<0.46	<0.48
Flatness of ends, FP2 (Limits ≤25μm)	46	42	45
Perpendicularity, P2 (Limits = Δ/L ≤0.0043)	<0.0028	<0.0032	<0.0033
Mean length, mm	176.00	168.00	170.00
Mean diameter, mm	75.00	74.00	75.00
Length to diameter ratio (L/D)	2.35	2.27	2.27
Moisture condition	as received	as received	as received
Moisture content, %	25.8	24.9	19.8
Visual description / Lithology	Gypsiferous SANDSTONE	Gypsiferous SANDSTONE	Gypsiferous SANDSTONE
Location of apparent weakness plane before test			
Failure pattern			

Sample temperature at test, °C	22.0	22	22
Rate of loading, mm/min	0.50	0.50	0.50
Time Taken for Failure (Min)	02:10	02:08	03:18
Unconfined compressive strength, MPa	1.97	1.95	4.15

*Notes: Length to diameter ratio <2.0 to 2.50

Method of sample preparation: ASTM D4543-19 Procedures /Straightness-S1, End Flatness FP2, Perpendicularity-P2

Test Method Variation: None

Remarks: Best effort achieved in End Flatness preparation

Test result relates only to the item tested.

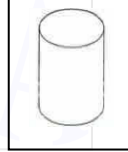
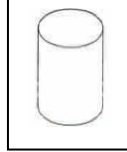
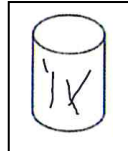
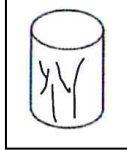
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Plate 26: Unconfined Compressive Strength for Borehole No. 5

UNCONFINED COMPRESSIVE STRENGTH TEST OF INTACT ROCK CORE REPORT

Date of sample received : 05th March 2023 Report Ref. No. : SIR2023-0018
Date of test started : 05th March 2023 Test method : ASTM D7012-14
Date of test finished : 11th September 2023 Sample prep. method : ASTM D4543-19
Plot No. – Location : JVC15AMRM004 – Al Barsha South Fourth. Tested by : SC.
Client : M/s. MDP Development Limited.

Source / Borehole No.	BH-5	BH-5
Sample No.	7	8
Depth, m	27.30 to 27.50	29.85 to 30.00

Straightness, S_1 (Limits $\leq 0.50\text{mm}$)	<0.41	<0.45
Flatness of ends, FP2 (Limits $\leq 25\mu\text{m}$)	46	42
Perpendicularity, P2 (Limits $= \Delta/L \leq 0.0043$)	<0.0030	<0.0032
Mean length, mm	174.00	111.00
Mean diameter, mm	75.00	75.00
Length to diameter ratio (L/D)	2.32	1.48*
Moisture condition	as received	as received
Moisture content, %	17.4	9.2
Visual description / Lithology	Conglomeratic SANDSTONE	CALCISILTITE
Location of apparent weakness plane before test		
Failure pattern		

Sample temperature at test, °C	22.0	22.0
Rate of loading, mm/min	0.5	0.5
Time Taken for Failure (Min)	03.42	04:12
Unconfined compressive strength, MPa	4.69	6.05

*Notes: Length to diameter ratio <2.0 to 2.50

Method of sample preparation: ASTM D4543-19 Procedures /Straightness-S1, End Flatness FP2, Perpendicularity-P2
Test Method Variation: None
Remarks: Best effort achieved in End Flatness preparation

Test result relates only to the item tested.

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Plate 27: Unconfined Compressive Strength for Borehole No. 5

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CHEMICAL TEST REPORT

Date of test started : 07th March 2023
Date of test finished : 11th March 2023
Report Ref. No. : SIR2023-0018
Plot No. – Location : JVC15AMRM004 – Al Barsha South Fourth, Dubai.
Client : M/s. MDP Development Limited.

Sampled by : HASA LAB
Sampled brought in by : HASA LAB
Sampling Method : BS 5930: 1999 Cl. 22
Tested by : VD
Checked by : MG

Borehole No.	Sample	Depth, m	pH Value	Chloride Content as Cl in acid soluble, %	Sulphate Content as So ₄ in 2 : 1 water/soil extract, g/l
Test method: BS 1377: Part 3: 1990			Cl: 9	Cl: 7.3	Cl: 5.3 & 5.5
2	Soil	1.00 : 1.15	8.4	0.04	0.44

Borehole No.	Sample	Depth, m	pH Value	Chloride Content as Cl, %	Sulphate Content as So ₄ , g/l
Test method: BS 1377: Part 3: 1990			Cl: 9	Cl: 7.2	Cl: 5.4 & 5.5
4	Groundwater	2.15	7.3	0.03	2.07

Test method variation: Commercial buffer solution used for pH meter calibration.

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Plate 28: Chemical Test Report

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APPENDIX D (CORE PHOTOS)





Plate 29: Borehole No. 1 (0.89m to 9.20m depth)



Plate 30: Borehole No. 1 (9.20m to 17.40m depth)



Plate 31: Borehole No. 1 17.40m to 26.00m depth)



Plate 32: Borehole No. 2 (26.00m to 30.00m depth)



Plate 33: Borehole No. 2 (1.87m to 10.50m depth)



Plate 34: Borehole No. 2 (10.50m to 18.75m depth)



Plate 35: Borehole No. 2 (18.75m to 27.00m depth)



Plate 36: Borehole No. 2 (27.00m to 36.30m depth)



Plate 37: Borehole No. 2 (36.30m to 40.00m depth)



Plate 38: Borehole No. 3 (0.86m to 10.00m depth)



Plate 39: Borehole No. 3 (10.00m to 18.50m depth)



Plate 40: Borehole No. 3 (18.50m to 26.50m depth)



Plate 41 Borehole No. 3 (26.50m to 30.00m depth)



Plate 42: Borehole No. 4 (0.85m to 9.80m depth)

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Plate 43: Borehole No. 4 (9.80m to 18.50m depth)

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Plate 44: Borehole No. 4 (18.50m to 26.70m depth)



Plate 45: Borehole No. 4 (26.70m to 30.00m depth)

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Plate 46: Borehole No. 5 (0.90m to 9.50m depth)

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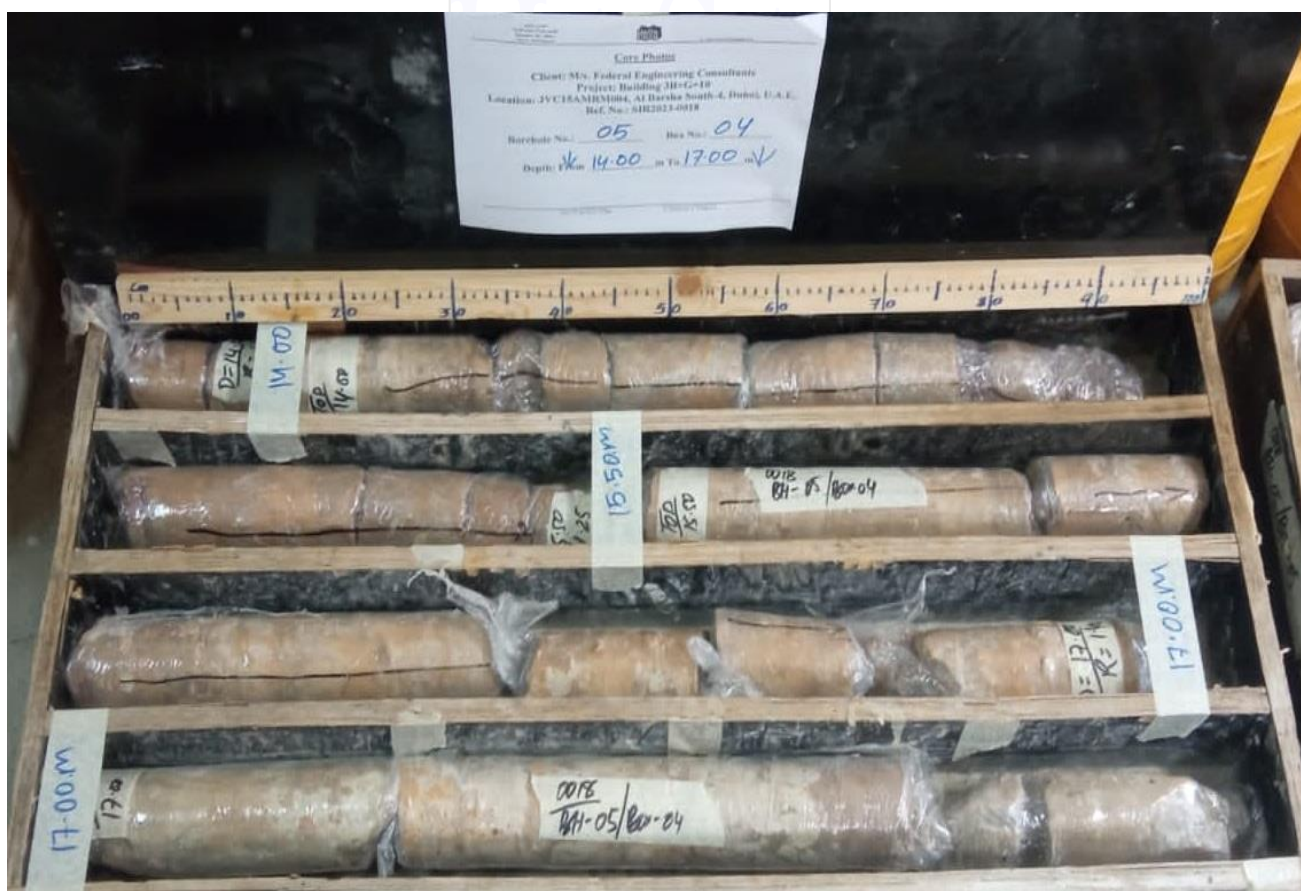


Plate 47: Borehole No. 5 (9.50m to 18.00m depth)

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Plate 48: Borehole No. 5 (18.00m to 27.50m depth)

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Plate 49: Borehole No. 5 (27.50m to 30.0m depth)

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APPENDIX E (USEFUL INFORMATION)

Percussion Boring Method

- The boring drilling machine consists of derrick, a power unit and winch carrying a light steel cable (wire) which passes through pulley on top of the derrick.
- The percussion tool is attached by the cable.
- The borehole is advanced by the percussive action of the tool, which is raised and dropped by means of the winch unit.
- The percussion tool is shell (bailer) for cohesion-less soils. It is heavy steel tube fitted with a flap or clack valve at the lower end.

Rotary Drilling Method

- Rotary drilling rig is operated by hydraulic system with (3 to 4-inches) cutting bit as drilling tool or core barrel with 3-inches diameter which is attached to the lower end of string of hollow drilling rods.
- The drilling tool may be either a cutting bit or a coring bit based on the subsurface conditions of the layers or the formation.
- The coring bit had been fixed at the lower end of a core barrel, which in turn is carried by drilling rods.
- Drilling mud/circulation fluid, which is a mix of water and bentonite pumped down through the hollow rods and passes under pressure generated by trash pump through narrow holes in the bit or barrel.
- The drilling fluid (mud) cools, lubricates the drilling rods and carries the loose debris (cuttings) to the surface through the annulus between the rods and the sides of the hole.
- The circulation fluid also introduces some supports to the sides of the holes when no casings are used.
- There are two forms of rotary drilling, open-hole drilling and core drilling. The details are as following:

- **Open-hole drilling**

The open-hole drilling method is generally used in soils, and very weak to weak rocks, uses cutting bit to break down all the material within the diameter of the hole. Open-hole drilling can be used, only, as a means of advancing the hole, the drilling rods can then be removed to allow tube samples to be taken or in-situ tests to be carried out.

- **Core drilling**

Core drilling is suitable for rocks and hard clays, the bit cuts an annular hole in the material and an intact core enters the barrel, to be removed as a sample. However, the natural water content of the material is liable to be increased due to contact with drilling fluid. Typical core diameters used are 2-inches and 3-inches.

Standard Penetration Test (SPT)

Standard Penetration Test was conducted according to BS 1377 part 9 : 1990 Cl. 3.3.

Standard Penetration Test (SPT) is performed in soil formation using split-barrel sampler. The sampler consists of a drive shoe at the bottom, a split barrel in the middle and a coupling at the top. The split barrel in the middle has inside and outside diameter of 35mm and 51mm respectively. The split-barrel sampler is attached to drive rod and then lowered to the bottom of the borehole. The sampler is driven into the soil by means of a hammer weighing 63.5kg and falls freely from a height of 760mm. The number of blows required for driving the sampler through three 150mm intervals is recorded. The sum of the number of blows required for driving the last two 150mm intervals is referred to as standard penetration number, N.

This test used mainly in sands and it can also be used in gravels or gravelly sand in which case the drive shoe may be replaced by a solid 60° cone, but when this accessory is used in any type of ground the result should be reported separately from standard test using open drive shoe and with preface SPT ©

Groundwater Table Measuring

The groundwater table was measured using Electric Contact Meter (Dip Meter), Type: KLL. Electric Contact Meters (Dip Meters) are a portable, used for quick, easy, reliable instruments for measuring the water level and total depth in boreholes, wells, observation pipes, reservoirs, as well as for the control of pumping tests. As soon as the electrode of the measuring probe touches the water surface, the signal lamp and on the instrument lights up and audible signal indicates. The water level can be read from the measuring tape in meters (m) and centimeters (cm). In most cases, the groundwater table level is confirmed by measuring the level using the traditional methods.

Particle Size Distribution

The Particle Size Distribution (Sieve Analysis) was conducted according to BS 1377 Part 2 : 1990: Cl. 9.2. and the classification of soil based on the particle sizes shown in the table below.

Classification	Size, mm
Clay	< 0.002
Silt	0.002 to 0.06
Fine Sand	0.06 to 0.20
Medium Sand	0.20 to 0.6
Coarse Sand	0.60 to 2.0
Fine Gravel	2.0 to 6.0
Medium Gravel	6.0 to 20.0
Coarse Gravel	20.0 to 60.0
Cobbles	60.0 to 200.0
Boulders	> 200.0

Unconfined Compressive Strength Test

Some of the samples extracted from the boreholes are core specimens. The unconfined compressive strength had been carried out on some selected core specimens using a calibrated proving ring (50kN, 10kN & 2kN capacity) with a load frame. The tests are carried out according to ASTM D 7012-10, ASTM D 4543-08. Two dial gauges were fitted as following: The first dial gauge is fitted in the load ring itself for measuring the load acting on the specimen and the second is for measuring the accompanied strain in the specimen.



SOIL CLASSIFICATION SYSTEM (Based on BS 5930:2015)

Table 9 Terms for classification of strength

Term based on measurement	Undrained strength classification definition, s_u , in kPa [from BS EN ISO 14688-2:2004, 5.3, Table 5]
Extremely low strength	<10
Very low strength	10 – 20
Low strength	20 – 40
Medium strength	40 – 75
High strength	75 – 150
Very high strength	150 – 300

Table 10 Terms for classification of relative density

Term	Classification based on uncorrected SPT N-values for use of borehole logs
Very loose	0 – 4
Loose	4 – 10
Medium dense	10 – 30
Dense	30 – 50
Very dense	>50

Figure 8 Plasticity chart

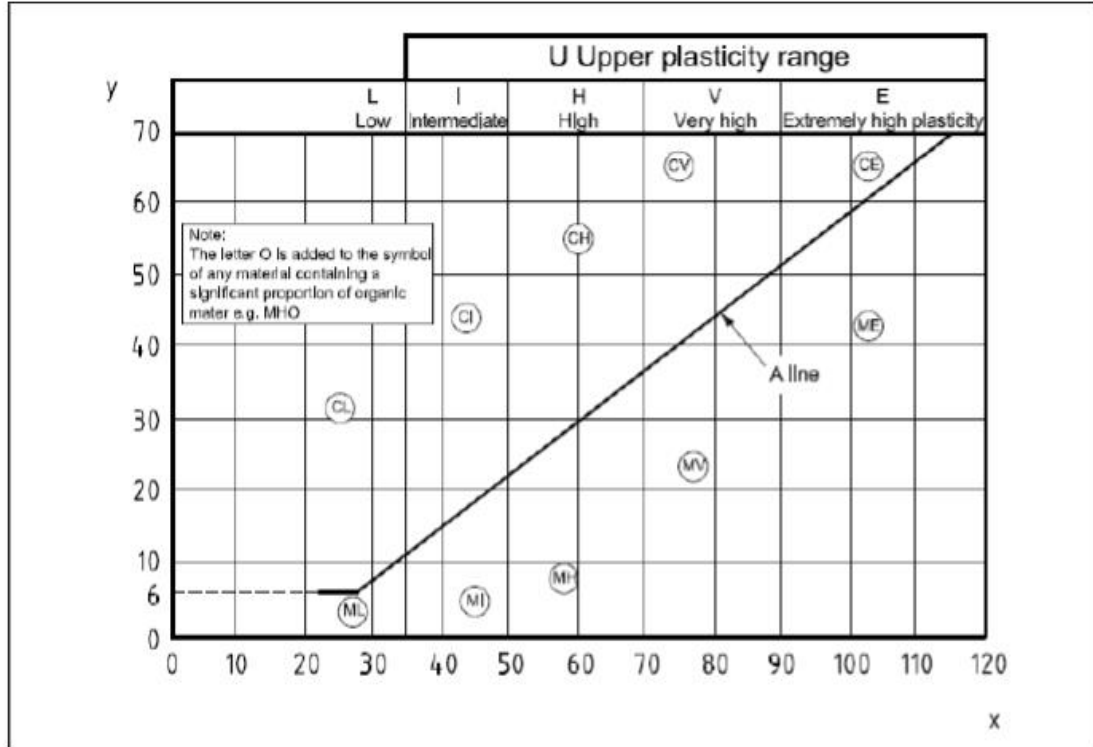


Table 17 Terms for mixtures of coarse and fine soils

Term	Principal soil type	Approximate proportion of secondary constituent	
		Coarse soil	Coarse and/or fine soil
slightly clayey or slightly silty and/or slightly sandy or slightly gravelly	SAND and/or GRAVEL	—	<5%
clayey or silty and/or sandy or gravelly			5% – 20% ^{A)}
very clayey or very silty and/or very sandy or very gravelly			>20% ^{A)}
very sandy or very gravelly	SILT ^{C)} or CLAY ^{C)}	>65% ^{B)}	—
sandy or gravelly		35% – 65%	
slightly sandy and/or slightly gravelly		<35%	

^{A)} Or described as fine soil depending on engineering behaviour.
^{B)} Or described as coarse soil depending on assessed engineering behaviour.
^{C)} Can be silty CLAY or clayey SILT.

ROCK CLASSIFICATION SYSTEM (Based on BS 5930:2015)

Table 25 Terms for description of rock strength

Term for use in field or based on measurement	Definition for field use	Definition on basis of Unconfined Compressive Strength measurements MPa
Extremely weak	Can be indented by thumbnail. Gravel sized lumps crush between finger and thumb.	0.6 – 1.0
Very weak	Crumbles under firm blows with point of geological hammer. Can be peeled by a pocket knife.	1 – 5
Weak	Can be peeled by a pocket knife with difficulty. Shallow indentations made by firm blow with the point of geological hammer.	5 – 25
Medium strong	Cannot be scraped with pocket knife. Can be fractured with a single firm blow of geological hammer.	25 – 50
Strong	Requires more than one blow of geological hammer to fracture.	50 – 100
Very strong	Requires many blows of geological hammer to fracture.	100 – 250
Extremely strong	Can only be chipped with geological hammer.	>250

NOTE Based on BS EN ISO 14689-1:2003 4.2.7, Table 5.

Table 26 Terms for description of thickness and spacing of structure

Thickness term	Spacing term	Thickness or spacing
Very thickly	Extremely wide	>6 m
Very thickly	Very wide	2 m – 6 m
Thickly	Wide	600 mm – 2 m
Medium	Medium	200 mm – 600 mm
Thinly	Close	60 mm – 200 mm
Very thinly	Very close	20 mm – 60 mm
Thickly laminated (Sedimentary)	Extremely close	6 mm – 20 mm
Narrowly (Metamorphic and Igneous)		
Thinly laminated (Sedimentary)	Extremely close	<6 mm
Very narrowly (Metamorphic and Igneous)		

NOTE A spacing of less than 20 mm or a bed thinner than 6 mm is still large in some deposits. Where finer features are observed, their spacing or thickness can also be given in mm, or additional terms used. For example, a partng is a bed which is only one or two grains thick.

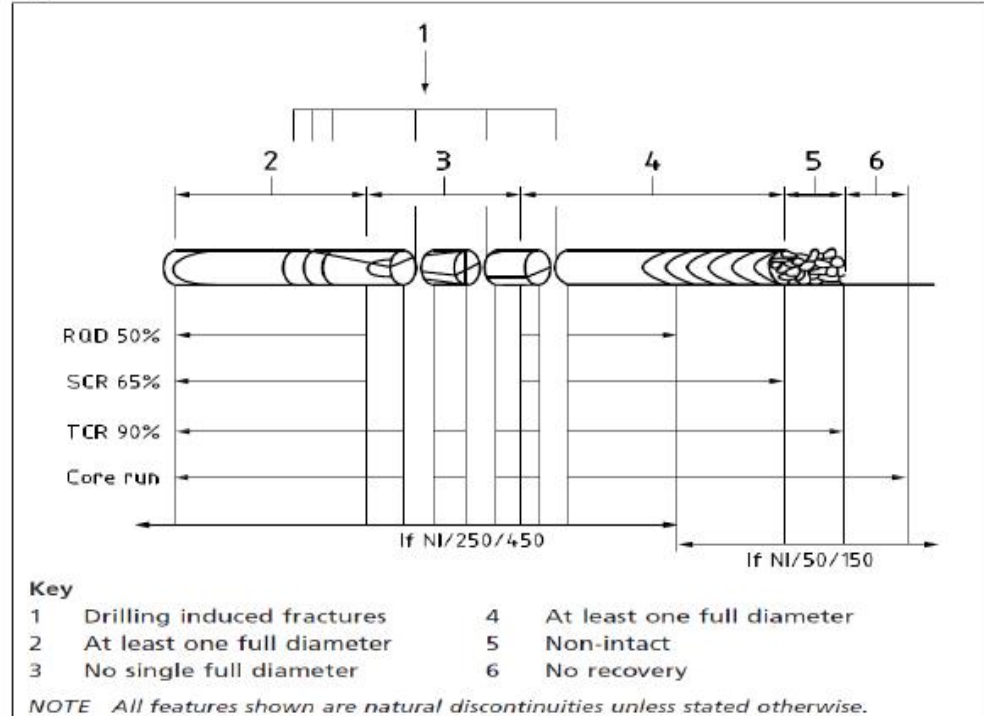
ROCK CLASSIFICATION SYSTEM (Based on BS 5930:2015)

Table 31 Terms for classification of discontinuity state (see Figure 10)

TCR (%)	Length of core recovered (solid and non-intact) expressed as a ratio of the length of core run.
SCR (%)	Length of solid core recovered expressed as a ratio of the length of core run. Solid core has a full diameter, uninterrupted by natural discontinuities, but not necessarily a full circumference and is commonly measured along the core axis or other scan line.
RQD (%)	Length of solid core each pieces longer than 100 mm expressed as a ratio of the length of core run.
Fracture index	Count of the number or spacing of fractures over an arbitrary length of core of similar intensity of fracturing recorded as minimum/mode/maximum. Commonly reported as Fracture Spacing (If, mm) or as Fracture Index (FI, number of fractures per metre). Where core is non-intact in the ground, the abbreviation NI may be used.

NOTE The total core recovery (TCR) records the proportion of core recovered and is read with the description, solid core recovery (SCR) and rock quality designation (RQD). The TCR of itself gives little information on the character of the core or the rock from which it was recovered. This measurement is required to ensure that all depth related records such as boundaries, markers and samples are correct.

Figure 10 Application of fracture state terms for rock cores



CIRIA Special Publication 31

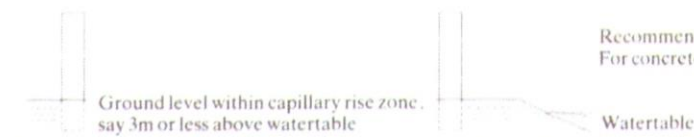
Figure 6 Typical exposure conditions

Note: These exposure conditions cover natural environments only. In industrial environments, concrete can be exposed to a range of more aggressive conditions, for example corrosive effluents, and may need to be specially protected.



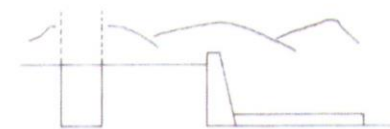
(a) Superstructures, inland with NO risk of windborne salts

Recommended cement type or
For concrete criteria, see Table 13



(b) Superstructures, in areas of saltflats, inland or near the coast, exposed to windborne salts

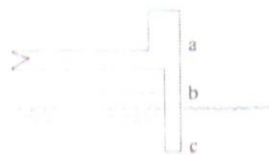
Recommended cement type or
For concrete criteria, see Table 13



(c) Parts of structures in contact with the soil, well above capillary rise zone, and with NO risk of water introduced at the surface by irrigation or faulty wastes, washing down, etc.

Recommended cement type or
For concrete criteria, see Table 13

Note: The recommendations for this exposure condition make NO PROVISION for protection from potentially aggressive soil conditions. Exposure conditions shown in Figure 6(d) should be assumed if it is believed that such protection might be needed



For superstructure: recommended cement types or
For concrete criteria, see Table 13

Recommended cement types	Exposure condition	For concrete criteria see Table 13
or	Splash zone	e(i)
or	Intertidal zone	e(ii)
or	Submerged zone	e(iii)

(e) Marine structures



For substructure: use recommendations given in Figure 6 (d)

(f) Water-retaining structures

CEMENT TYPE

Where resistance is needed against **sulphate** attack, but there is NO significant risks of chloride-induced corrosion, **SULPHATE-RESISTING CEMENT** to BS 4027 or ASTM Type V (i.e. cement with a maximum C_3A content of 3.5 or 5.0%, respectively gives better protection.

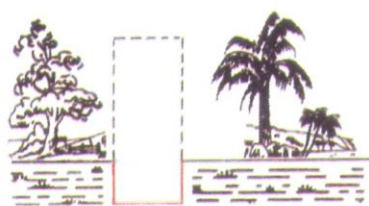
Colour key:

Where improved resistance is needed against **chloride induced corrosion** of the reinforcement, but there is NO significant exposure to sulphates. Cement with a medium to high C_3A content is preferred. OPC or ASTM Type I usually have high C_3A contents and ASTM Type II usually has a medium C_3A content (but those specifications do not specify minimum C_3A contents. See Section 8.2.8).

Colour key:

Where resistance is needed against **both sulphates and chlorides**, concrete may need to be protected from the soil and groundwater with a waterproof membrane or tanking, and a compromise has to be made on the type of cement used. Generally, a cement containing at least 3.5% but not more than 9% C_3A is preferred. Each situation should be considered on its merits. See also Section 8, which gives more information on different types of cement. Colour key:

THE IMPERMEABILITY OF THE CONCRETE HAS MUCH MORE INFLUENCE THAN THE CEMENT TYPE, UNDER ALL CONDITIONS OF EXPOSURE



Surfaces from which water evaporates are coloured blue. Under conditions d(ii) to (iv) of Table 13, salts are likely to become concentrated in these areas and extra protection may be needed. Surfaces where protection with waterproof membranes or tanking is advised under conditions of significant contamination, are coloured red (see Table 13)

Reinforcement corrosion is unlikely to be rapid in concrete which is permanently saturated: cement type ■ is recommended for exposures d(ii) and d(iv) in this situation

Recommended cement types	Soil and groundwater conditions ⁽¹⁾	For concrete criteria, see Table 13
■ or ■ ■ or ■	Free from significant contamination ⁽²⁾ Significant sulphate contamination only Significant chloride contamination only Significant contamination with both sulphates and chlorides	d(i) d(ii) d(iii) d(iv)

Notes:

(1) Fill and made-up ground within capillary rise zone will reach the same level of contamination as the surrounding soil.

(2) BRE Digest 250 and CP 110 grade soils and groundwater in five steps of sulphate concentration: 0.2% total sulphate or 1.0g/l in 2:1 soil water extract is considered significant. There is no widely accepted view on the concentration at which chlorides become significant in soil or groundwater, but limited experience in the Gulf region suggests it may be as low as 0.05%, particularly in situations where alternate wetting and drying or capillary rise affect the concrete.

(d) Parts of structures in contact with soil within the capillary rise zone, below groundwater level, or where water may be introduced at the surface by irrigation, discharge of wastes, washing down, etc.

Table 13 Current local criteria for reinforced concrete in the Gulf region

Specifications currently in use in the Gulf region for a wide range of building and civil engineering works agree reasonably well on ranges for minimum cement content, maximum water/cement ratio, and minimum cover to reinforcement. Ranges of values typical of these specifications are given here for the information of the user who needs to specify limits appropriate to the particular environment, exposure, and operational conditions. Lack of long-term experience precludes making specific recommendations in this Guide.

EXPOSURE CONDITIONS INDICATED IN FIGURE 6(a) to (f) (Local specifications do not distinguish between all of these conditions.)	RANGE OF SPECIFICATION LIMITS			
	Minimum cement content for 20 mm aggregates (kg/m ³) ⁽¹⁾	Maximum water/cement ratio ⁽¹⁾	Additional requirements ⁽¹⁾	Minimum cover for reinforcement (mm)
a	300 to 320	0.52 to 0.50	NONE	30
b	320	0.50	NONE	40
c (i) to (iv)	320 to 350	0.50 to 0.45	NONE	40 to 50
d (i)	300 to 320	0.50	NONE	40 to 50
(ii) or (iv)	320 to 400 ⁽²⁾	0.50 to 0.42 ⁽²⁾	Tanking/membrane ⁽²⁾	40 to 50
(iii)	Not separately mentioned in specifications: treat as d (ii) to (iv)			
e (i) to (iii)	370 to 400	0.45 to 0.42	NONE	75 to 100
f	400 ⁽³⁾	0.50 ⁽³⁾	NONE ⁽³⁾	40 ⁽³⁾

Notes:

1 Criteria for unreinforced concrete do not differ substantially from these requirements.

2 The wide range of these requirements reflects the range of sulphate concentrations in the soil or groundwater, but takes no account of chloride concentration. (The five levels of significant sulphate concentration adopted in BRE Digest 250 and CP 110 are used in local specifications.) (see References 28 and 29).

3 This information is taken from one local specification only.

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CARBONATE CLASSIFICATION SYSTEM (Based on Clark & Walker)

INCREASING GRAIN SIZE OF PARTICULATE DEPOSITS		TOTAL CARBONATE CONTENT %	
80mm	2mm	0.06mm	0.002mm
Carbonate GRAVEL	Carbonate SAND	Carbonate SILT	Carbonate MUD (carb. clay)
Siliceous Carbonate GRAVEL	Siliceous Carbonate SAND	Siliceous Carbonate SILT	Clayey carbonate MUD (Carbonate clay)
Calcareous Silica GRAVEL	Calcareous Silica SAND	Calcareous Silica SILT	Calcareous CLAY
Silica GRAVEL	Silica SAND	Silica SILT	CLAY
CALCIRUDITE (carb. congl. or breccia)	CALCARENITE (Calc. Sandst.)	CALCISILTITE (Carb. Siltstone)	CALCILTITE (carb. mudstone)
Siliceous CALCIRUDITE	Siliceous CALCARENITE	Siliceous CALCISILTITE	Clayey CALCILTITE
Calcareous CONGLOMERATE	Calcareous SANDSTONE	Calcareous SILTSTONE	Calcareous CLAYSTONE
CONGLOMERATE or BRECCIA	SANDSTONE	SILTSTONE	CLAYSTONE
	Detrital LIMESTONE	Fine grained LIMESTONE	Fine grained Argillaceous LIMESTONE
	Siliceous Detrital LIMESTONE	Siliceous LIMESTONE	Calcareous CLAYSTONE
	Calcareous CONGLOMERATE	Calcareous SILTSTONE	CLAYSTONE
	CONGLOMERATE or BRECCIA	SILTSTONE	
	CRYSTALLINE LIMESTONE or MARBLE		
	Conventional metamorphic nomenclature applies in this section		

INCREASING LITHIFICATION

	SOIL to WEAK ROCK		ROCK	
Rock Strength	$\sigma_c < 5 \text{ MPa}$ Very weak to weak	$\sigma_c 5 - 12.5 \text{ MPa}$ Moderately weak	$\sigma_c 12.5 - 100 \text{ MPa}$ Moderately strong to strong	$\sigma_c > 100 \text{ MPa}$ Very strong to extremely strong
Cementation Degree	Non cemented to Moderately cemented	Well cemented	(Well cemented) Increasing cementation decreasing porosity	Re-crystallised
Cone Resistance	$q_c < 35 \text{ MPa}$	$q_c 35 - 85 \text{ MPa}$	$q_c > 85 \text{ MPa}$	

SITE CLASSIFICATION AND GROUND MOTION PARAMETERS (ASCE-7-16)

Table 20.3-1 Site Classification

Site Class	$\bar{V}_s$	$\bar{N}$ or $\bar{N}_{ch}$	$\bar{S}_u$
A. Hard rock	>5,000 ft/s	NA	NA
B. Rock	2,500 to 5,000 ft/s	NA	NA
C. Very dense soil and soft rock	1,200 to 2,500 ft/s	>50 blows/ft	>2,000 lb/ft ²
D. Stiff soil	600 to 1,200 ft/s	15 to 50 blows/ft	1,000 to 2,000 lb/ft ²
E. Soft clay soil	<600 ft/s	<15 blows/ft	<1,000 lb/ft ²
Any profile with more than 10 ft of soil that has the following characteristics:			
— Plasticity index $PI > 20$,			
— Moisture content $w \geq 40\%$,			
— Undrained shear strength $\bar{s}_u < 500$ lb/ft ²			
F. Soils requiring site response analysis in accordance with Section 21.1	See Section 20.3.1		

Note: For SI: 1 ft = 0.3048 m; 1 ft/s = 0.3048 m/s; 1 lb/ft² = 0.0479 kN/m².

Table 11.4-1 Short-Period Site Coefficient, F_s

Mapped Risk-Targeted Maximum Considered Earthquake (MCE _R) Spectral Response Acceleration Parameter at Short Period						
Site Class	$S_s \leq 0.25$	$S_s = 0.5$	$S_s = 0.75$	$S_s = 1.0$	$S_s = 1.25$	$S_s \geq 1.5$
A	0.8	0.8	0.8	0.8	0.8	0.8
B	0.9	0.9	0.9	0.9	0.9	0.9
C	1.3	1.3	1.2	1.2	1.2	1.2
D	1.6	1.4	1.2	1.1	1.0	1.0
E	2.4	1.7	1.3	See Section 11.4.8	See Section 11.4.8	See Section 11.4.8
F	See Section 11.4.8	See Section 11.4.8	See Section 11.4.8	See Section 11.4.8	See Section 11.4.8	See Section 11.4.8

Note: Use straight-line interpolation for intermediate values of S_s .

Table 11.4-2 Long-Period Site Coefficient, F_v

Mapped Risk-Targeted Maximum Considered Earthquake (MCE _R) Spectral Response Acceleration Parameter at 1-s Period						
Site Class	$S_1 \leq 0.1$	$S_1 = 0.2$	$S_1 = 0.3$	$S_1 = 0.4$	$S_1 = 0.5$	$S_1 \geq 0.6$
A	0.8	0.8	0.8	0.8	0.8	0.8
B	0.8	0.8	0.8	0.8	0.8	0.8
C	1.5	1.5	1.5	1.5	1.5	1.4
D	2.4	2.2 ^a	2.0 ^a	1.9 ^a	1.8 ^a	1.7 ^a
E	4.2	See Section 11.4.8	See Section 11.4.8	See Section 11.4.8	See Section 11.4.8	See Section 11.4.8
F	See Section 11.4.8	See Section 11.4.8	See Section 11.4.8	See Section 11.4.8	See Section 11.4.8	See Section 11.4.8

Note: Use straight-line interpolation for intermediate values of S_1 .

^aAlso, see requirements for site-specific ground motions in Section 11.4.8.

TABLE 11.8-1 Site Coefficient F_{PGA}

Mapped Maximum Considered Geometric Mean (MCE _G) Peak Ground Acceleration, PGA						
Site Class	$PGA \leq 0.1$	$PGA = 0.2$	$PGA = 0.3$	$PGA = 0.4$	$PGA = 0.5$	$PGA \geq 0.6$
A	0.8	0.8	0.8	0.8	0.8	0.8
B	0.9	0.9	0.9	0.9	0.9	0.9
C	1.3	1.2	1.2	1.2	1.2	1.2
D	1.6	1.4	1.3	1.2	1.1	1.1
E	2.4	1.9	1.6	1.4	1.2	1.1
F	See Section 11.4.8					

Note: Use straight-line interpolation for intermediate values of PGA.

Table 1.5-2 Importance Factors by Risk Category of Buildings and Other Structures for Snow, Ice, and Earthquake Loads

Risk Category from Table 1.5-1	Snow Importance Factor, I_s	Ice Importance Factor—Thickness, I_t	Ice Importance Factor—Wind, I_w	Seismic Importance Factor, I_e
I	0.80	0.80	1.00	1.00
II	1.00	1.00	1.00	1.00
III	1.10	1.15	1.00	1.25
IV	1.20	1.25	1.00	1.50

Note: The component importance factor, I_p , applicable to earthquake loads, is not included in this table because it depends on the importance of the individual component rather than that of the building as a whole, or its occupancy. Refer to Section 13.1.3.

LIST OF ABBREVIATIONS USED IN THE REPORT*

ASTM	: American Society of Testing and Materials
BGL	: Below Ground Level
BH	: Borehole
BS	: British Standard
CPT	: Cone Penetration Test
Dr	: Relative Density
Dia.	: Diameter
EGSL	: Existing Ground Surface Level
Gs	: Specific Gravity
HASA	: Hassan Al Amir Soil Analysis
K _a	: Coefficient of Active Earth Pressure
K _o	: Coefficient of Earth Pressure at Rest
K _p	: Coefficient of Passive Earth Pressure
k _s	: Modulus of Subgrade Reaction
LL	: Liquid Limit
m	: Mass
NA	: Not Applicable
PL	: Plastic Limit
PSD	: Particle Size Distribution
SL	: Shrinkage Limit
SPT	: Standard Penetration Test
SPT ©	: SPT with Solid Cone
TBM	: Temporary Bench Mark
Wc	: Water Content
γ	: Bulk Density
γ_{dry}	: Dry Density
γ_{Sat}	: Saturated Density
ϕ	: Angle of Shearing Resistance
τ	: Shear Strength of Soil

For borehole log :

D	: Disturbed Sample
U	: Undisturbed Sample
B	: Bulk Disturbed Sample
S	: SPT with split spoon sampler
SC	: SPT with solid cone
C	: Core Run
GW	: Groundwater
GS	: Ground Surface
N	: Standard Penetration Test Number
UCS	: Unconfined Compressive Strength
TCR	: Total Core Recovery, %
SCR	: Solid Core Recovery, %
RQD	: Rock Quality Designation, %

* Some of above abbreviations may not encountered in the report

LIST OF SYMBOLS USED IN THE REPORT*

SYMBOL LEGEND FOR GEOTECHNICAL LOGS

SOIL TYPES:

MADE GROUND			
BOULDERS & COBBLES			
GRAVEL	Poorly graded	SILT	High plasticity
	Well graded		Low plasticity
SAND	Poorly graded	CLAY	High plasticity
	Well graded		Low plasticity
		PEAT/ ORGANIC CLAY	
Composite soil types:	Signified by combined symbols; eg. Silty SAND		
Additional Secondary Terms:	Shelly	Gypsiferous	

ROCK TYPES:

Sedimentary:

CLAYSTONE/ MUDSTONE	
SILTSTONE	
SANDSTONE	
CONGLOMERATE	
BRECCIA	
LIMESTONE	
CHALK	

Carb. CLAYSTONE/ CALCILUTITE	
Carb. SILTSTONE/ CALCISILTITE	
Carb. SANDSTONE/ CALCARENITE	
Carb. CONGLOM./ CALCIRUDITE	

Evaporite:

GYPSUM/ HALITE	
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Igneous:

Coarse Grained	
Medium Grained	
Fine Grained	

Metamorphic:

Coarse Grained	
Medium Grained	
Fine Grained	

SAMPLE TYPES:

Bulk/ disturbed	Shelby/ undisturbed	Standard Penetration Test (SPT)	No Recovery (SPT)	Rock Core
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*Some of above symbols may not encountered in the report

REFERENCES

Karl Terzaghi & Ralph B. Peck.

SOIL MECHANICS IN ENGINEERING PRACTICE, 3rd Edition.

Braja M. Das

PRINCIPLES OF GEOTECHNICAL ENGINEERING, 5th Edition.

Joseph E. Bowles

FOUNDATION ANALYSIS AND DESIGN, 5th Edition.

MJ Tomlinson

FOUNDATION DESIGN AND CONSTRUCTION, 6th Edition.

BS 5930: 2015.

CODE OF PRACTICE FOR SITE INVESTIGATION.

CIRIA SPECIAL PUBLICATION-31.

The CIRIA Guide to Concrete Construction in the Gulf Region, published by the Construction Industry Research and Information Association, London 1984.

ion, London 1984.